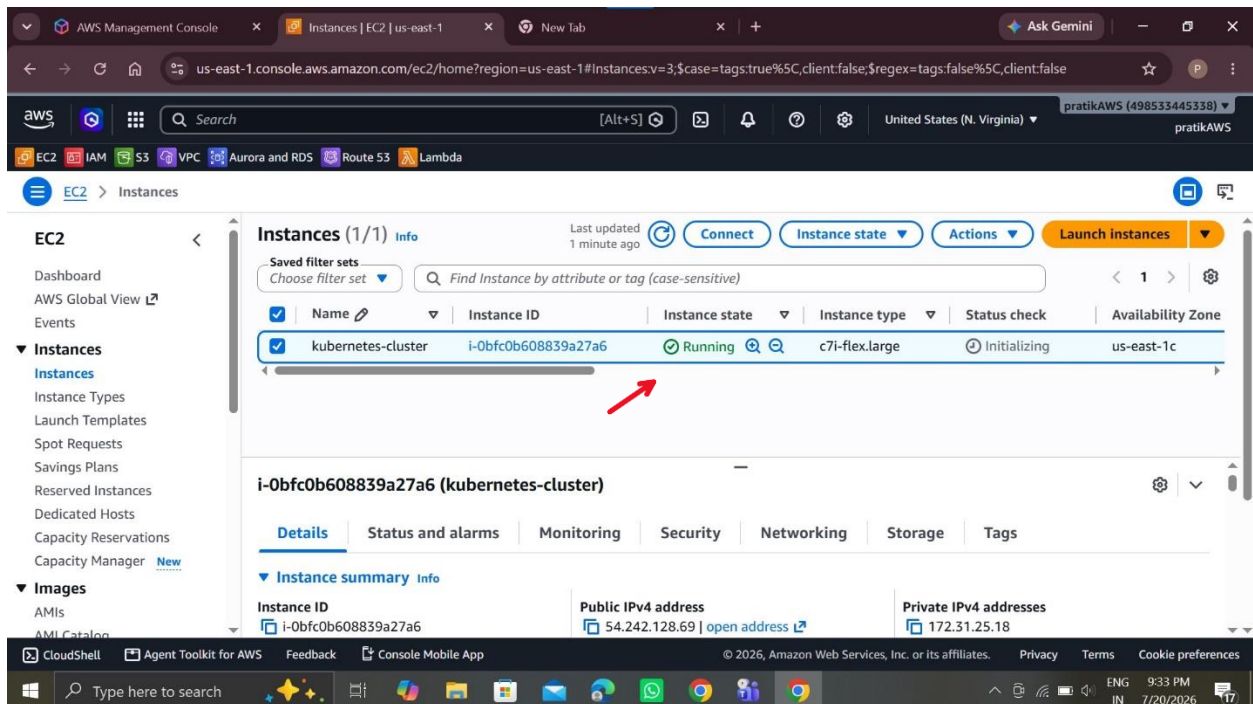
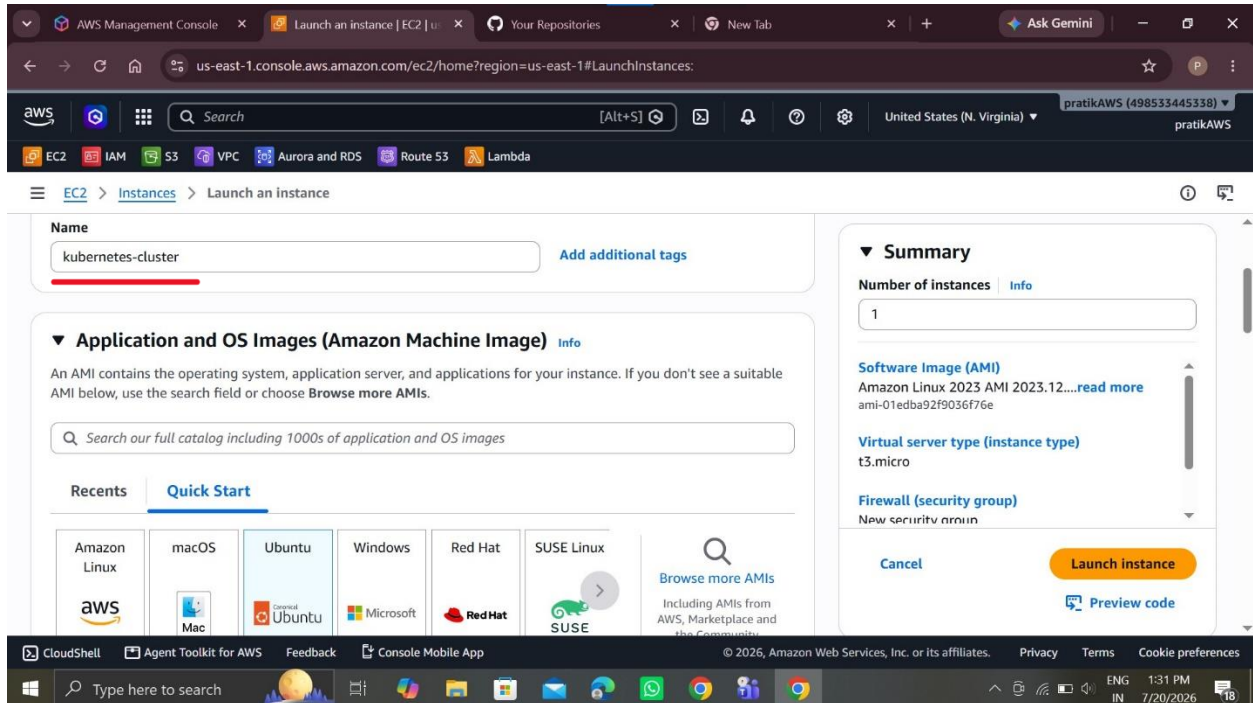


K8s Kind Voting App Deployment and configure argocd

Launch Kubernetes cluster



SSH through Login Kubernetes cluster in Git Bash Terminal

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ConnectToInstance:instanceId=i-0bfc0b608839a27a6

EC2 > Instances > i-0bfc0b608839a27a6 > Connect to instance

EC2 Instance Connect | SSM Session Manager | **SSH client** | EC2 serial console

Instance ID
i-0bfc0b608839a27a6 (kubernetes-cluster)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is devops-live-k8s-project.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "devops-live-k8s-project.pem"`
4. Connect to your instance using its Public DNS:
28-69.compute-1.amazonaws.com

✓ Command copied

```
ssh -i "devops-live-k8s-project.pem" ubuntu@ec2-54-242-128-69.compute-1.amazonaws.com
```

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

CloudShell | Agent Toolkit for AWS | Feedback | Console Mobile App | © 2025, Amazon Web Services, Inc. or its affiliates. | Privacy | Terms | Cookie preferences

Update system

```
ubuntu@ip-172-31-25-18: ~$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [137 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [136 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [16.3 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [6329 kB]
Get:6 http://security.ubuntu.com/ubuntu InRelease [137 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [137 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [313 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [291 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [127 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [50.0 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [8276 B]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [358 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [91.7 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [53.1 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [5756 B]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [209 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [65.0 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [153 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [4704 B]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [255 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [47.6 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [3360 B]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [772 B]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [216 B]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [256 B]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [212 B]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [112 B]
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [216 B]
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [116 B]
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [216 B]
Get:32 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [120 B]
Get:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease [216 B]
```

Install docker

```
ubuntu@ip-172-31-25-18: ~$  
Get:40 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [44.2 kB]  
Get:41 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]  
Get:42 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]  
Get:43 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [248 kB]  
Get:44 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [46.5 kB]  
Get:45 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]  
Get:46 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]  
Fetched 30.5 MB in 3s (10.9 MB/s)  
Reading package lists... Done  
ubuntu@ip-172-31-25-18:~$ sudo apt-get install docker.io  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Solving dependencies... Done  
The following additional packages will be installed:  
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan  
Suggested packages:  
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse  
  rootlesskit zfs-fuse | zfsutils  
The following NEW packages will be installed:  
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan  
0 upgraded, 8 newly installed, 0 to remove and 70 not upgraded.  
Need to get 74.0 MB of archives.  
After this operation, 281 MB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 pigz amd64 2.8-1build1 [68.5 kB]  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 bridge-utils amd64 1.7.1-4ubuntu3 [34.8 kB]  
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]  
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 containerd amd64 2.2.2-0ubuntu1.1 [28.1 MB]  
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 dns-root-data all 2025080400build1 [6022 B]  
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 dnsmasq-base amd64 2.92-1ubuntu0.4 [441 kB]  
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 docker.io amd64 29.1.3-0ubuntu4 .1 [35.5 MB]  
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 ubuntu-fan all 0.12.17 [34.3 kB]  
Fetched 74.0 MB in 1s (105 MB/s)  
Preconfiguring packages ...  
Selecting previously unselected package pigz.  
(Reading database ... 85303 files and directories currently installed.)
```

Modify Docker USER permission

```
ubuntu@ip-172-31-25-18: ~$  
Selecting previously unselected package ubuntu-fan.  
Preparing to unpack .../7-ubuntu-fan_0.12.17_all.deb ...  
Unpacking ubuntu-fan (0.12.17) ...  
Setting up dnsmasq-base (2.92-1ubuntu0.4) ...  
Setting up runc (1.4.0-0ubuntu1) ...  
Setting up dns-root-data (2025080400build1) ...  
Setting up bridge-utils (1.7.1-4ubuntu3) ...  
Setting up pigz (2.8-1build1) ...  
Setting up containerd (2.2.2-0ubuntu1.1) ...  
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' -> '/usr/lib/systemd/system/containerd.service'.  
Setting up ubuntu-fan (0.12.17) ...  
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' -> '/usr/lib/systemd/system/ubuntu-fan.service'.  
Setting up docker.io (29.1.3-0ubuntu4.1) ...  
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' -> '/usr/lib/systemd/system/docker.service'.  
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' -> '/usr/lib/systemd/system/docker.socket'.  
Processing triggers for dbus (1.16.2-2ubuntu4) ...  
Processing triggers for man-db (2.13.1-1build1) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-25-18:~$  
ubuntu@ip-172-31-25-18:~$ docker ps  
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock  
ubuntu@ip-172-31-25-18:~$ sudo usermod -s /bin/bash $(cat /etc/passwd | grep docker | cut -d ':' -f 1) && newgrp docker  
ubuntu@ip-172-31-25-18:~$ docker ps  
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES  
ubuntu@ip-172-31-25-18:~$  
ubuntu@ip-172-31-25-18:~$  
ubuntu@ip-172-31-25-18:~$
```


Install kind

```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~$
ubuntu@ip-172-31-25-18:~$ mkdir k8s-install
ubuntu@ip-172-31-25-18:~$ ls
k8s-install
ubuntu@ip-172-31-25-18:~$ cd k8s-install/
ubuntu@ip-172-31-25-18:~/k8s-install$ vim install_kind.sh
ubuntu@ip-172-31-25-18:~/k8s-install$ vim install_kind.sh
ubuntu@ip-172-31-25-18:~/k8s-install$ vim install_kind.sh
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ chmod +x install_kind.sh
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ ./install_kind.sh
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100    97    100    97    0      0    2171      0      0     0
0      0      0      0      0      0      0      0     0
100  6.15M    100  6.15M    0      0   37.75M      0      0     0
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kind --version
kind version 0.20.0
ubuntu@ip-172-31-25-18:~/k8s-install$ |
```

Configure kind cluster & install kubectl

```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$ vim config.yml
ubuntu@ip-172-31-25-18:~/k8s-install$ kind create cluster --config=config.yml --name=my-cluster
Creating cluster "my-cluster" ...
 ✓ Ensuring node image (kindest/node:v1.30.0)
 ✓ Preparing nodes
 ✓ Writing configuration
 ✓ Starting control-plane
 ✓ Installing CNI
 ✓ Installing StorageClass
 ✓ Joining worker nodes
Set kubectl context to "kind-my-cluster"
You can now use your cluster with:

kubectl cluster-info --context kind-my-cluster

Thanks for using kind! ☺
ubuntu@ip-172-31-25-18:~/k8s-install$ vim install_kubectl.sh
ubuntu@ip-172-31-25-18:~/k8s-install$ chmod +x install_kubectl.sh
ubuntu@ip-172-31-25-18:~/k8s-install$ ./install_kubectl.sh
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 49.07M    100 49.07M    0      0   157.0M      0      0     0
Client Version: v1.30.0
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
kubectl installation complete.
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
my-cluster-control-plane            Ready    control-plane   3m35s   v1.30.0
my-cluster-worker                   Ready    <none>         3m13s   v1.30.0
my-cluster-worker2                  Ready    <none>         3m14s   v1.30.0
ubuntu@ip-172-31-25-18:~/k8s-install$
```

Create argocd namespace

```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get namespace
NAME                STATUS    AGE
default             Active   4m48s
kube-node-lease     Active   4m48s
kube-public         Active   4m48s
kube-system         Active   4m48s
local-path-storage  Active   4m44s
ubuntu@ip-172-31-25-18:~/k8s-install$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                    NAMES
1f445762f6ba   kindest/node:v1.30.0 "/usr/local/bin/entr..." 5 minutes ago  Up 5 minutes  127.0.0.1:33593->6443/tcp  my-cluster-control-plan
4f3c8162ea62   kindest/node:v1.30.0 "/usr/local/bin/entr..." 5 minutes ago  Up 5 minutes                        my-cluster-worker
94743bfcb201   kindest/node:v1.30.0 "/usr/local/bin/entr..." 5 minutes ago  Up 5 minutes                        my-cluster-worker2
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl create namespace argocd
namespace/argocd created
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml
customresourcedefinition.apiextensions.k8s.io/applications.argoproj.io created
customresourcedefinition.apiextensions.k8s.io/appprojects.argoproj.io created
serviceaccount/argocd-application-controller created
serviceaccount/argocd-applicationset-controller created
serviceaccount/argocd-dex-server created
serviceaccount/argocd-notifications-controller created
serviceaccount/argocd-repo-server created
serviceaccount/argocd-server created
role.rbac.authorization.k8s.io/argocd-application-controller created
role.rbac.authorization.k8s.io/argocd-applicationset-controller created
role.rbac.authorization.k8s.io/argocd-dex-server created
role.rbac.authorization.k8s.io/argocd-notifications-controller created
role.rbac.authorization.k8s.io/argocd-repo-server created
role.rbac.authorization.k8s.io/argocd-server created
clusterrole.rbac.authorization.k8s.io/argocd-application-controller created
clusterrole.rbac.authorization.k8s.io/argocd-applicationset-controller created
clusterrole.rbac.authorization.k8s.io/argocd-server created
rolebinding.rbac.authorization.k8s.io/argocd-application-controller created
rolebinding.rbac.authorization.k8s.io/argocd-applicationset-controller created
rolebinding.rbac.authorization.k8s.io/argocd-dex-server created
```

Successfully show argocd pods

```
ubuntu@ip-172-31-25-18: ~/k8s-install
configmap/argocd-notifications-cm created
configmap/argocd-rbac-cm created
configmap/argocd-ssh-known-hosts-cm created
configmap/argocd-tls-certs-cm created
secret/argocd-notifications-secret created
secret/argocd-secret created
service/argocd-applicationset-controller created
service/argocd-dex-server created
service/argocd-metrics created
service/argocd-notifications-controller-metrics created
service/argocd-repo-server created
service/argocd-server created
service/argocd-server-metrics created
deployment.apps/argocd-applicationset-controller created
deployment.apps/argocd-dex-server created
deployment.apps/argocd-notifications-controller created
deployment.apps/argocd-repo-server created
deployment.apps/argocd-server created
statefulset.apps/argocd-application-controller created
networkpolicy.networking.k8s.io/argocd-application-controller-network-policy created
networkpolicy.networking.k8s.io/argocd-applicationset-controller-network-policy created
networkpolicy.networking.k8s.io/argocd-dex-server-network-policy created
networkpolicy.networking.k8s.io/argocd-notifications-controller-network-policy created
networkpolicy.networking.k8s.io/argocd-repo-server-network-policy created
networkpolicy.networking.k8s.io/argocd-server-network-policy created
The CustomResourceDefinition "applicationssets.argoproj.io" is invalid: metadata.annotations: Too long: must have at most 262144 bytes
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods -n argocd
NAME                                READY   STATUS    RESTARTS   AGE
argocd-application-controller-0     1/1     Running   0           56s
argocd-applicationset-controller-7b8ccc5477-mxzmg 1/1     Running   0           57s
argocd-dex-server-64f9f64b68-grjj7 1/1     Running   0           57s
argocd-notifications-controller-8656595f88-nknb4 1/1     Running   0           57s
argocd-repo-server-59db66d7d4-2tm8k 1/1     Running   0           57s
argocd-repo-server-5864959f88-xgqch 1/1     Running   0           57s
argocd-server-564847b6b7-lmhxr     1/1     Running   0           57s
ubuntu@ip-172-31-25-18:~/k8s-install$
```

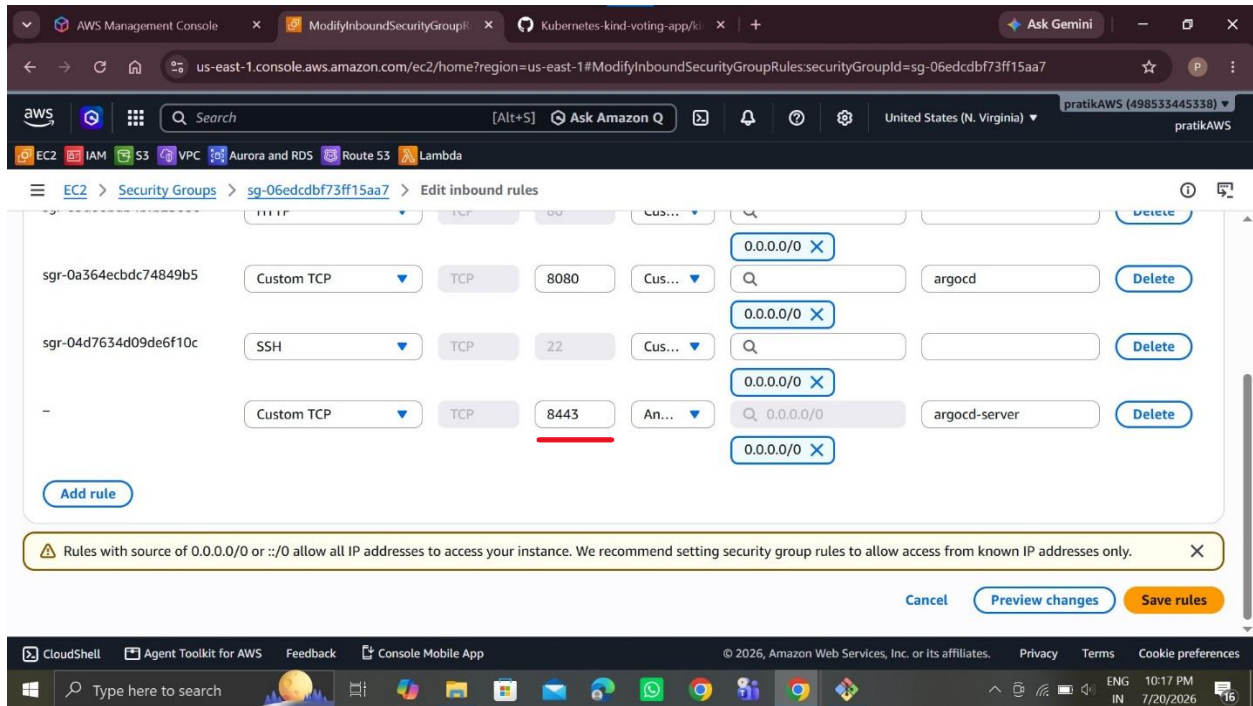
Then patch argocd server in Node port

```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get svc -n argocd
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)                                     AGE
argocd-applicationset-controller    ClusterIP           10.96.188.185    <none>           7000/TCP,8080/TCP                         2m18s
argocd-dex-server                   ClusterIP           10.96.77.224     <none>           5556/TCP,5557/TCP,5558/TCP               2m18s
argocd-metrics                      ClusterIP           10.96.91.23      <none>           8082/TCP                                   2m18s
argocd-notifications-controller-metrics ClusterIP           10.96.151.64     <none>           9001/TCP                                   2m18s
argocd-redis                       ClusterIP           10.96.219.96     <none>           6379/TCP                                   2m18s
argocd-repo-server                  ClusterIP           10.96.144.107    <none>           8081/TCP,8084/TCP                        2m18s
argocd-server                       ClusterIP           10.96.230.184    <none>           80/TCP,443/TCP                           2m18s
argocd-server-metrics              ClusterIP           10.96.107.60     <none>           8083/TCP                                   2m18s
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl patch svc argocd-server -n argocd -p '{"spec": {"type": "NodePort"}}'
service/argocd-server patched
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get svc -n argocd
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)                                     AGE
argocd-applicationset-controller    ClusterIP           10.96.188.185    <none>           7000/TCP,8080/TCP                         3m26s
argocd-dex-server                   ClusterIP           10.96.77.224     <none>           5556/TCP,5557/TCP,5558/TCP               3m26s
argocd-metrics                      ClusterIP           10.96.91.23      <none>           8082/TCP                                   3m26s
argocd-notifications-controller-metrics ClusterIP           10.96.151.64     <none>           9001/TCP                                   3m26s
argocd-redis                       ClusterIP           10.96.219.96     <none>           6379/TCP                                   3m26s
argocd-repo-server                  ClusterIP           10.96.144.107    <none>           8081/TCP,8084/TCP                        3m26s
argocd-server                       NodePort            10.96.230.184    <none>           80:32645/TCP,443:30940/TCP              3m26s
argocd-server-metrics              ClusterIP           10.96.107.60     <none>           8083/TCP                                   3m26s
ubuntu@ip-172-31-25-18:~/k8s-install$
```

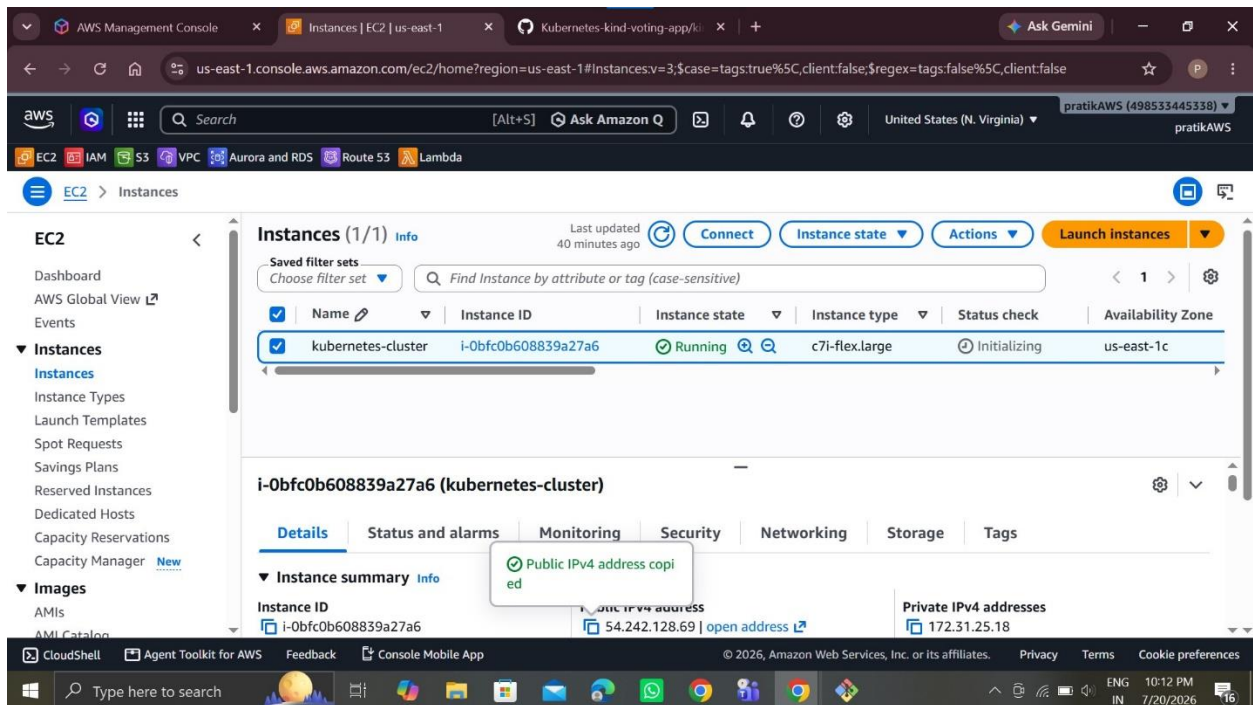
Port forward in argocd server

```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get svc -n argocd
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)                                     AGE
argocd-applicationset-controller    ClusterIP           10.96.188.185    <none>           7000/TCP,8080/TCP                         5m13s
argocd-dex-server                   ClusterIP           10.96.77.224     <none>           5556/TCP,5557/TCP,5558/TCP               5m13s
argocd-metrics                      ClusterIP           10.96.91.23      <none>           8082/TCP                                   5m13s
argocd-notifications-controller-metrics ClusterIP           10.96.151.64     <none>           9001/TCP                                   5m13s
argocd-redis                       ClusterIP           10.96.219.96     <none>           6379/TCP                                   5m13s
argocd-repo-server                  ClusterIP           10.96.144.107    <none>           8081/TCP,8084/TCP                        5m13s
argocd-server                       NodePort            10.96.230.184    <none>           80:32645/TCP,443:30940/TCP              5m13s
argocd-server-metrics              ClusterIP           10.96.107.60     <none>           8083/TCP                                   5m13s
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl port-forward svc/argocd-server -n argocd 30940:443 --address=0.0.0.0 &
[1] 8785
ubuntu@ip-172-31-25-18:~/k8s-install$ Forwarding from 0.0.0.0:30940 -> 8080
^C
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl port-forward svc/argocd-server -n argocd 8443:443 --address=0.0.0.0 &
[2] 9143
ubuntu@ip-172-31-25-18:~/k8s-install$ Forwarding from 0.0.0.0:8443 -> 8080
```

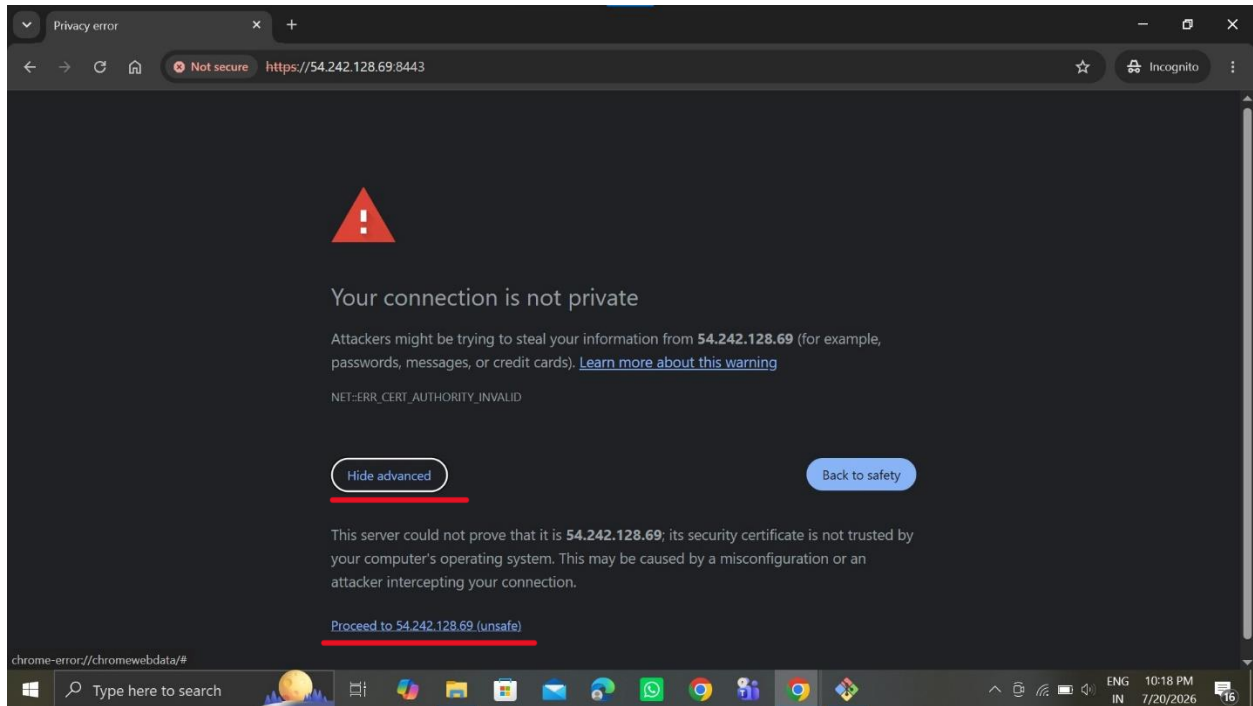

Allow this port in Security group



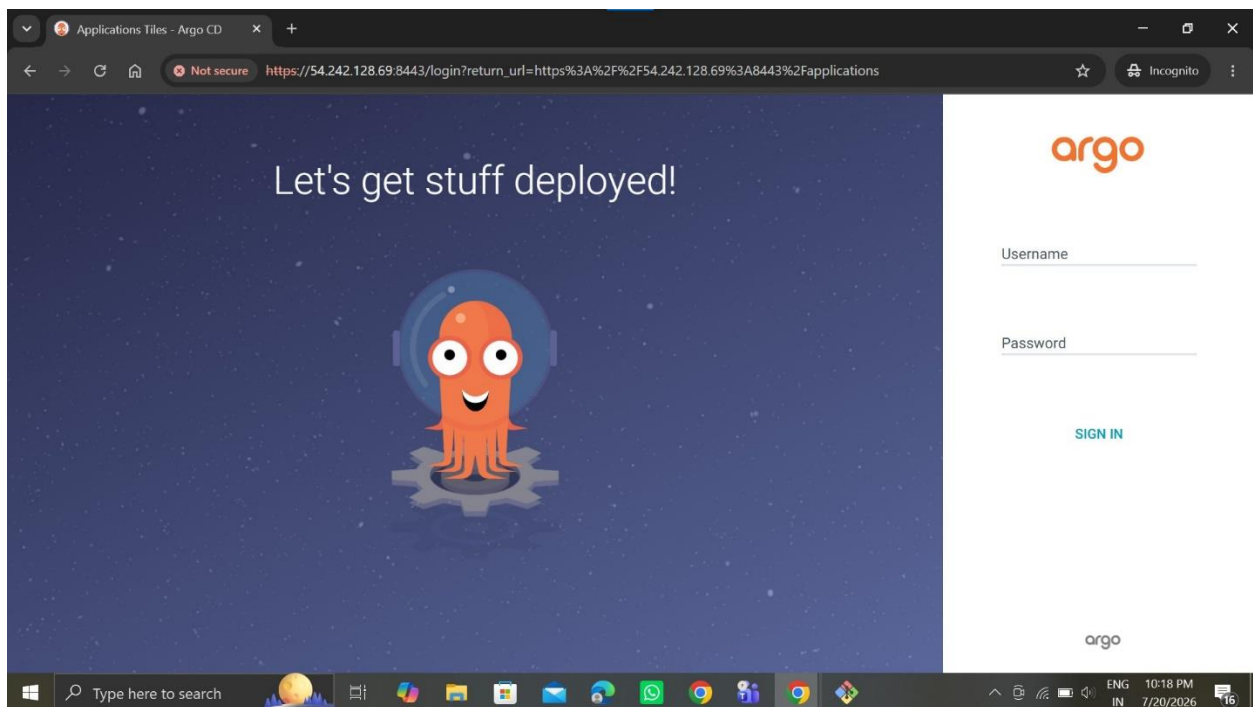
Then copy Public IP of Kubernetes cluster & go to browser



Paste this public IP & click Proceed



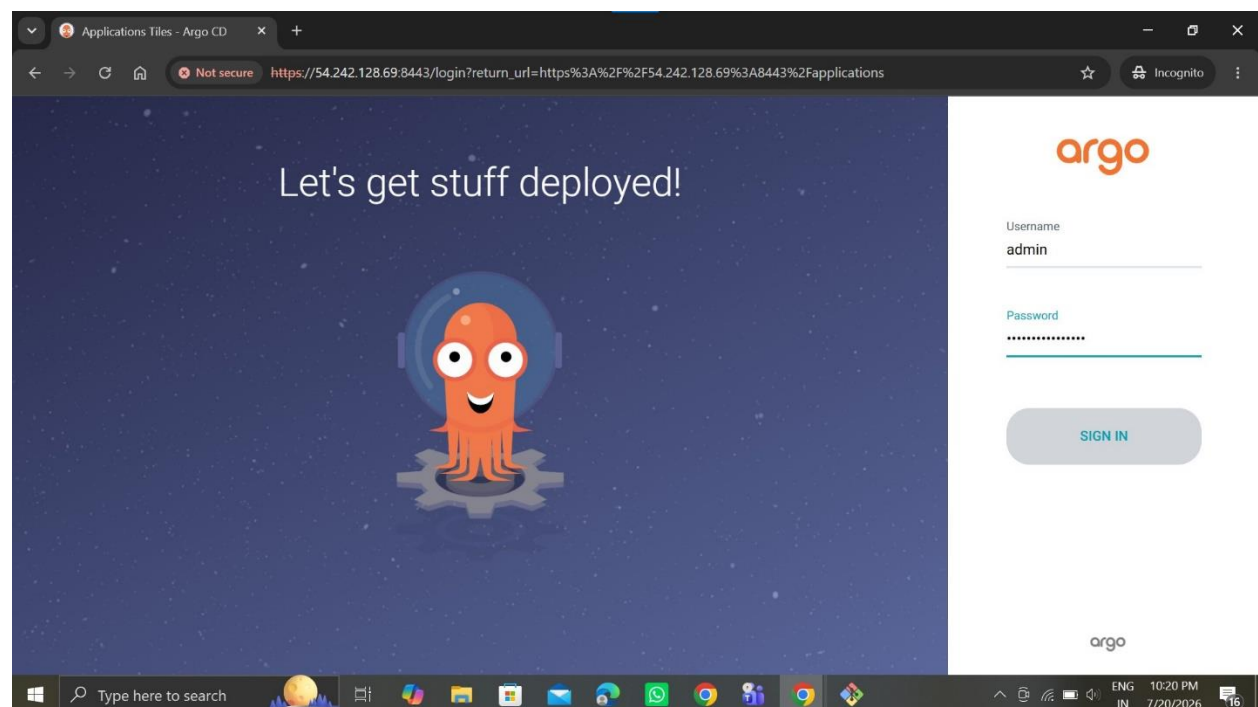
Successfully show the argocd login page



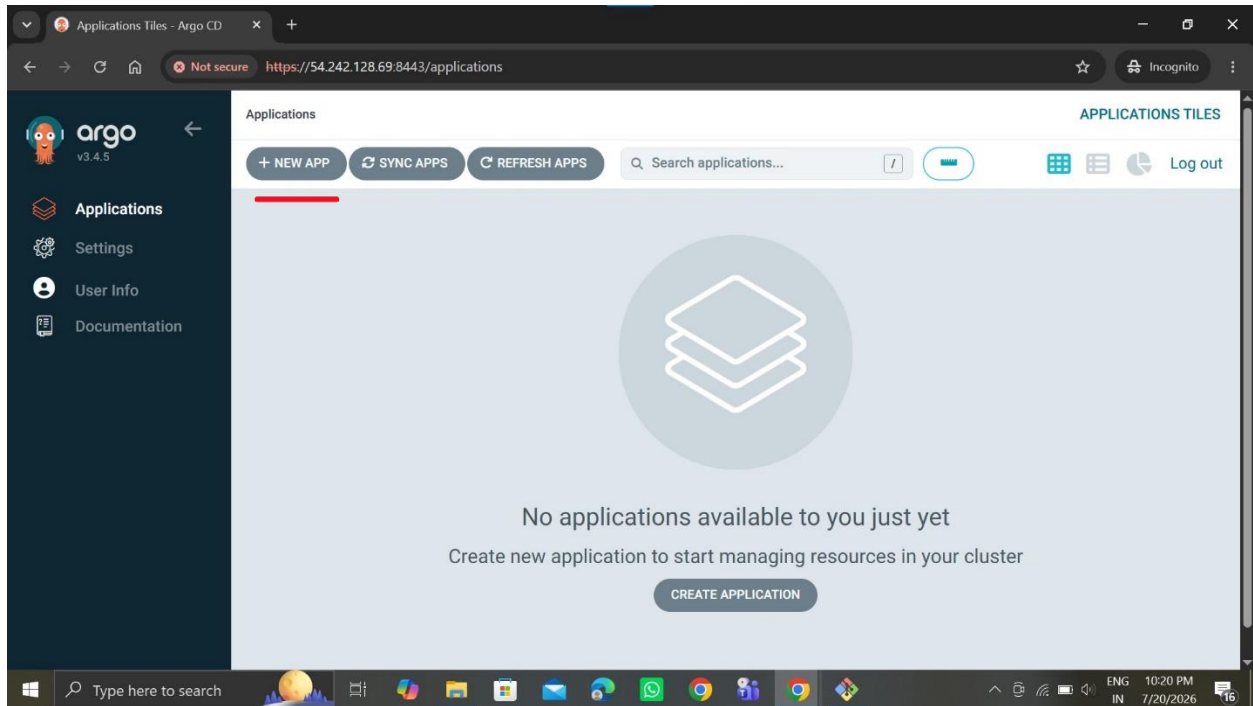
Then generate argocd initial admin password

[illegible]

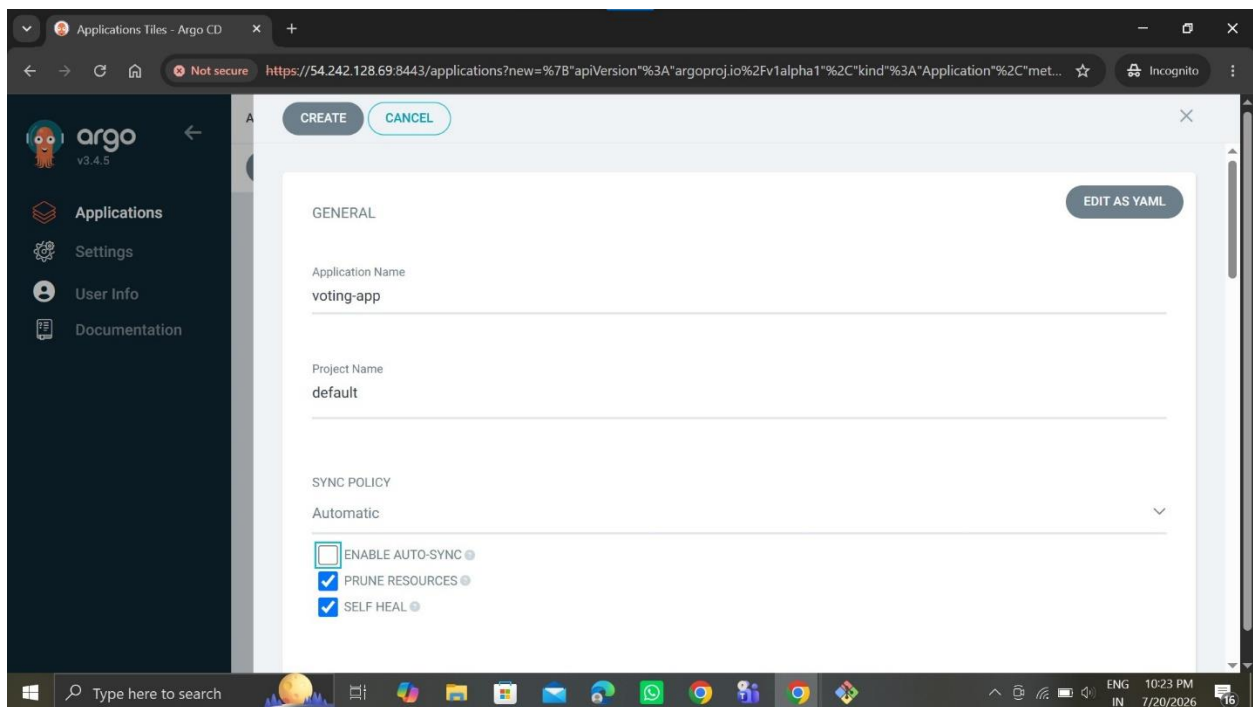
Then give the user name , password & then sign in



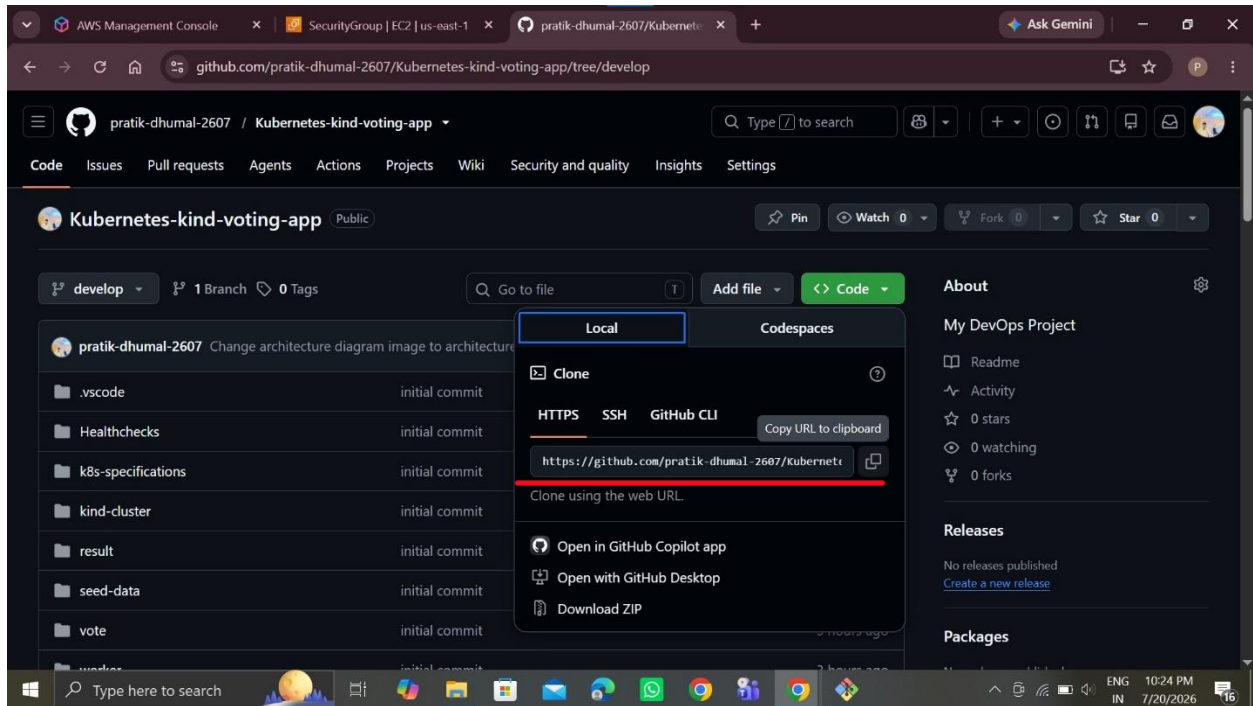
Successfully show argocd dashboard then click New App



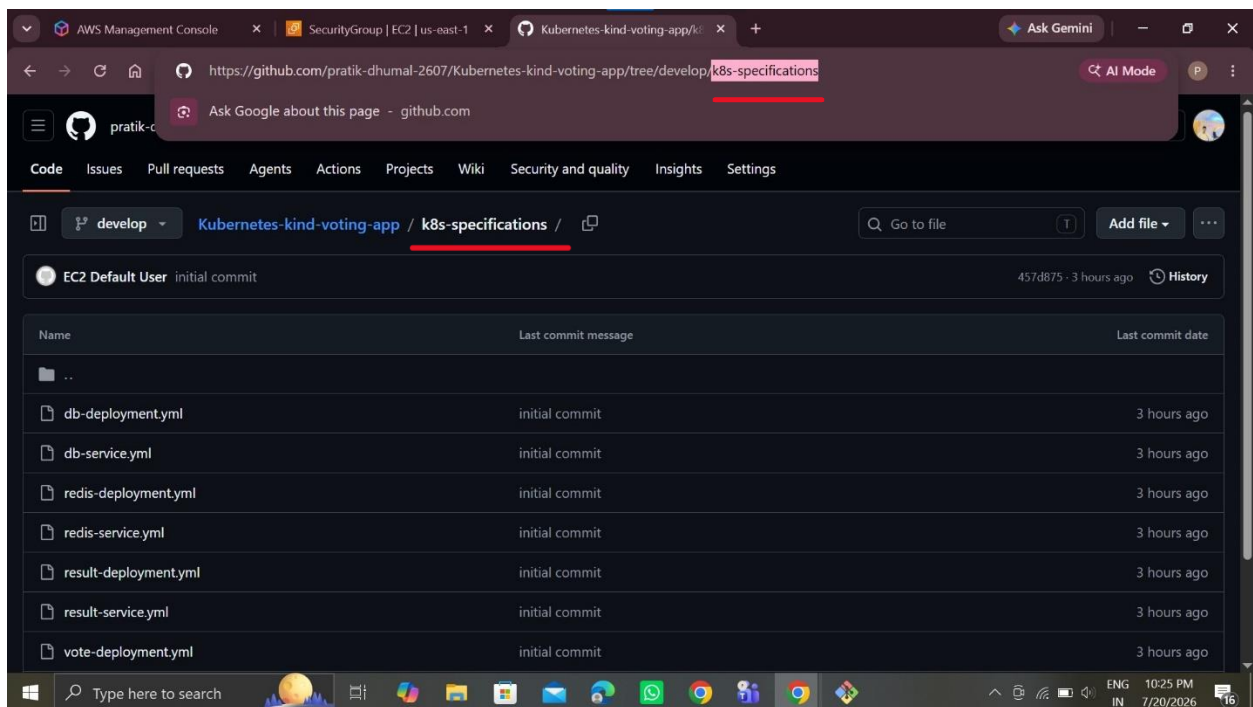
Fill application info



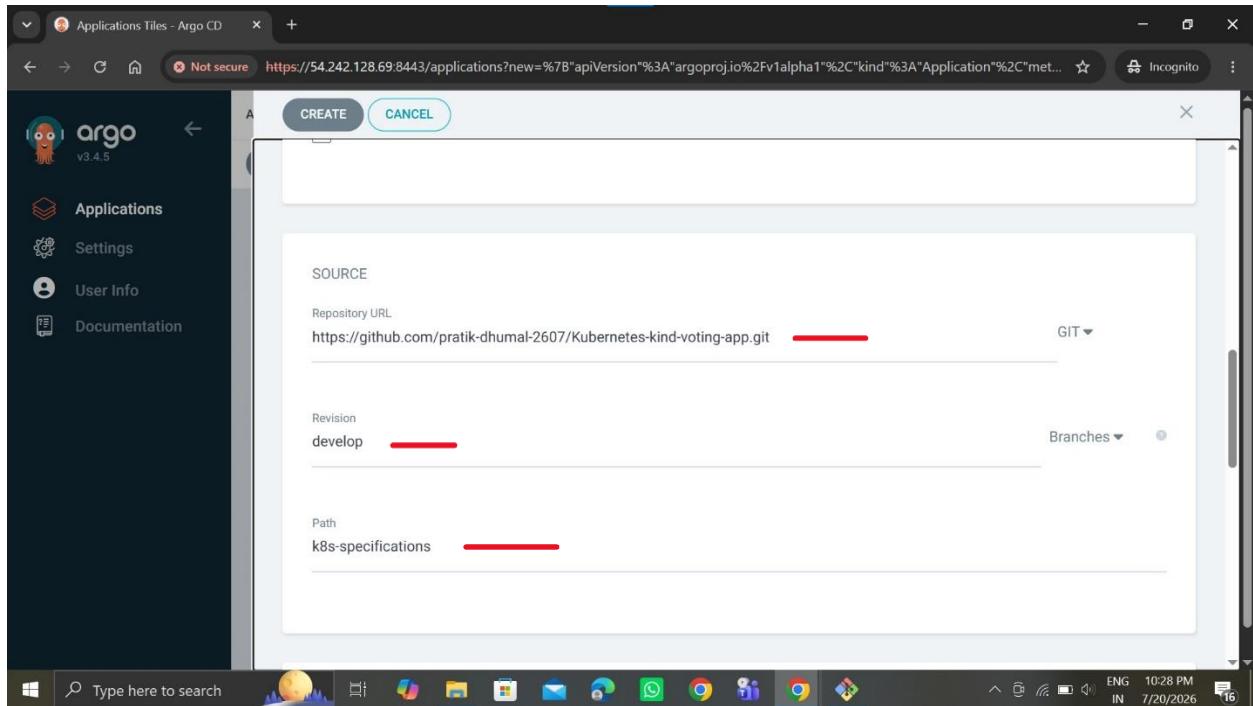
Copy github repository URL



Copy the Path of manifest files folder



Paste this repository URL , select repository branch , paste path of manifest files folder

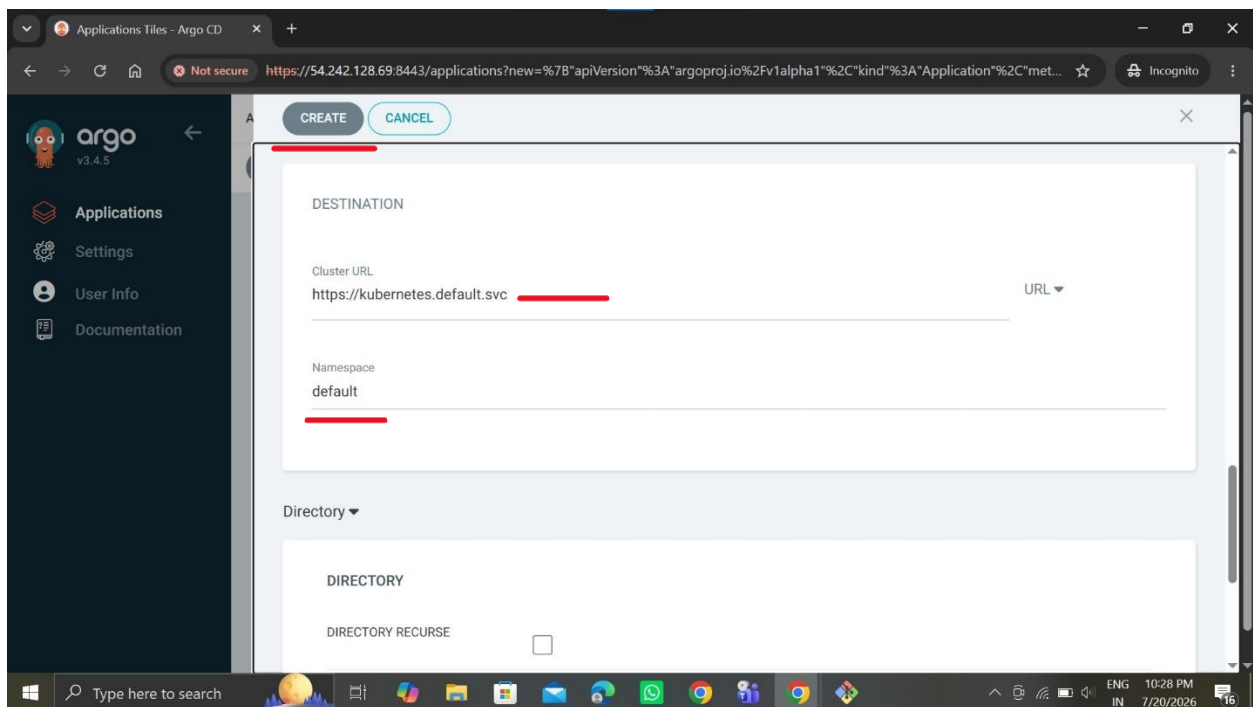


The screenshot shows the Argo CD web interface with the 'CREATE' form open. The 'SOURCE' section is visible, containing the following fields:

- Repository URL:** `https://github.com/pratik-dhumal-2607/Kubernetes-kind-voting-app.git`
- Revision:** `develop`
- Path:** `k8s-specifications`

The form also includes a 'GIT' dropdown menu and a 'Branches' dropdown menu. The left sidebar shows the Argo CD logo and navigation links: Applications, Settings, User Info, and Documentation.

Then set cluster URL & set namespace default then click Create

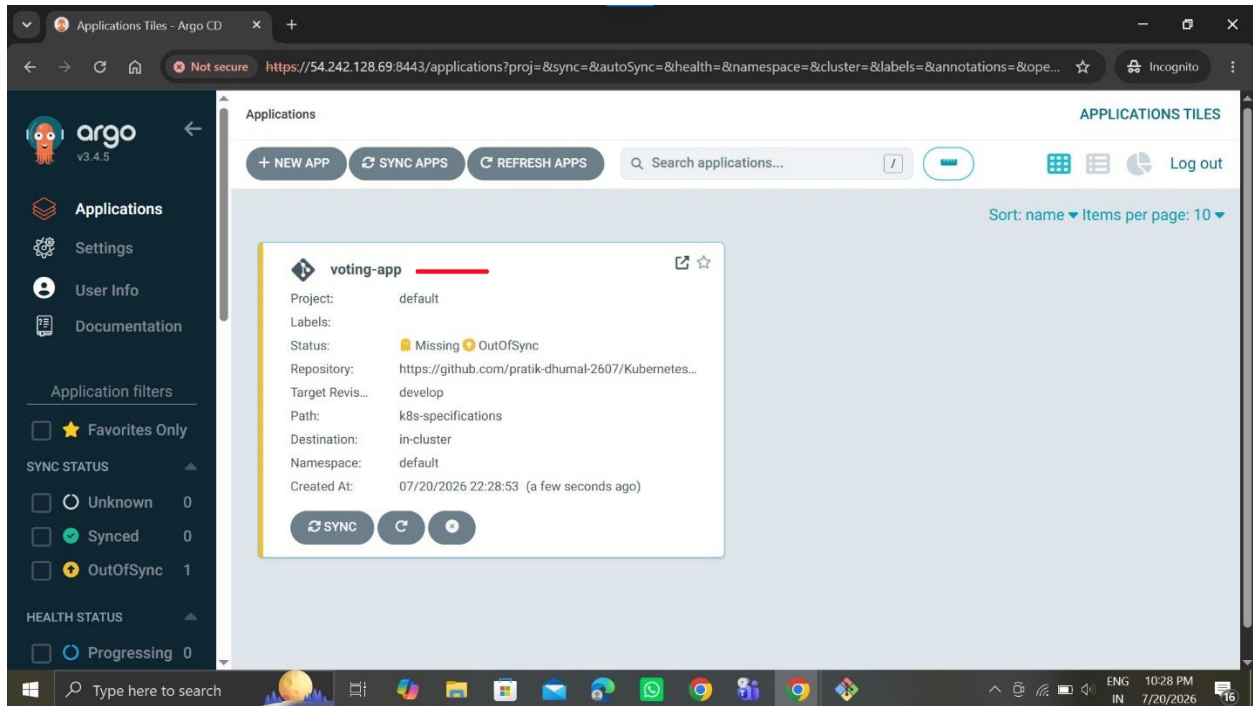


The screenshot shows the Argo CD web interface with the 'CREATE' form open. The 'DESTINATION' section is visible, containing the following fields:

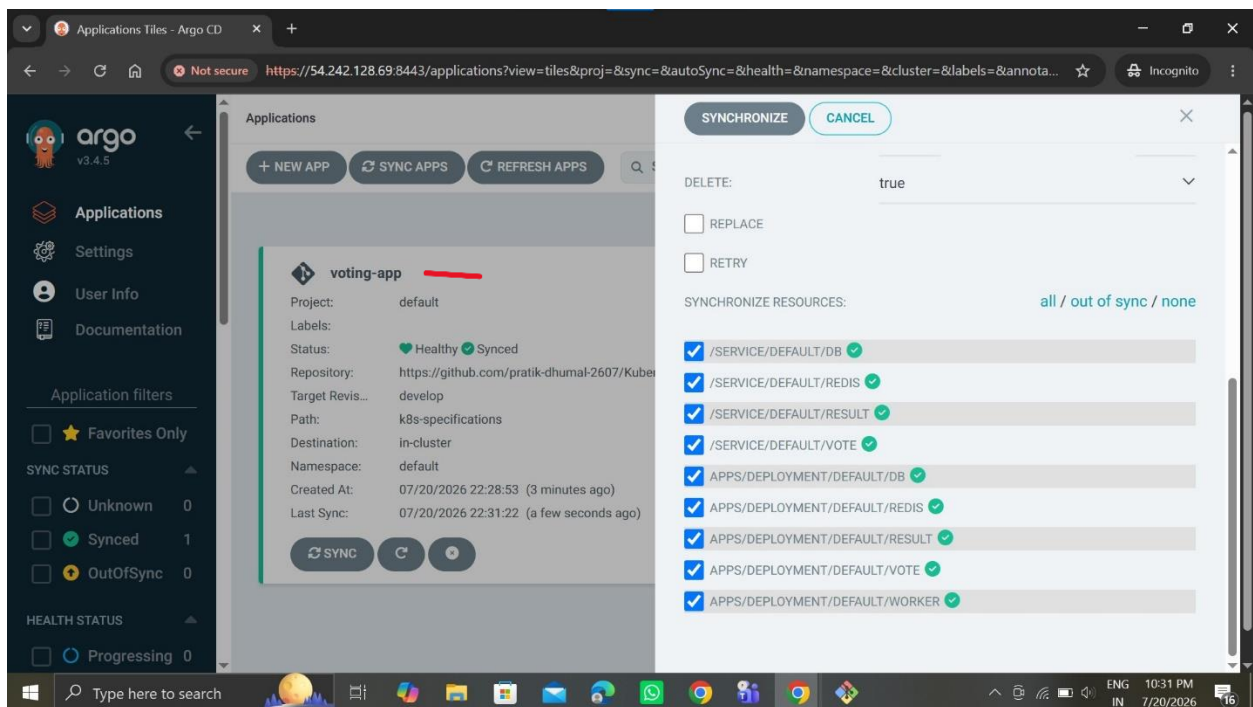
- Cluster URL:** `https://kubernetes.default.svc`
- Namespace:** `default`

Below the 'DESTINATION' section, there is a 'Directory' dropdown menu and a 'DIRECTORY RECURSE' checkbox. The left sidebar shows the Argo CD logo and navigation links: Applications, Settings, User Info, and Documentation.

Application successfully created



Then click this application



Successfully run this application

The screenshot shows the Argo CD web interface in a browser window. The URL is `https://54.242.128.69:8443/applications/argocd/voting-app?resource=&view=tree`. The left sidebar contains the Argo logo (v3.4.5) and navigation links: Applications, Settings, User Info, and Documentation. Below these are resource filters for NAME, KINDS, and SYNC STATUS. The main content area is titled 'APPLICATION DETAILS TREE' and includes tabs for DETAILS, DIFF, SYNC, SYNC STATUS, HISTORY AND ROLLBACK, DELETE, and REFRESH. The application status is 'Healthy' and 'Synced to develop (945b96b)'. The last sync was successful 8 minutes ago. The main view displays a tree diagram of the application's resources, including pods, services, and deployments.

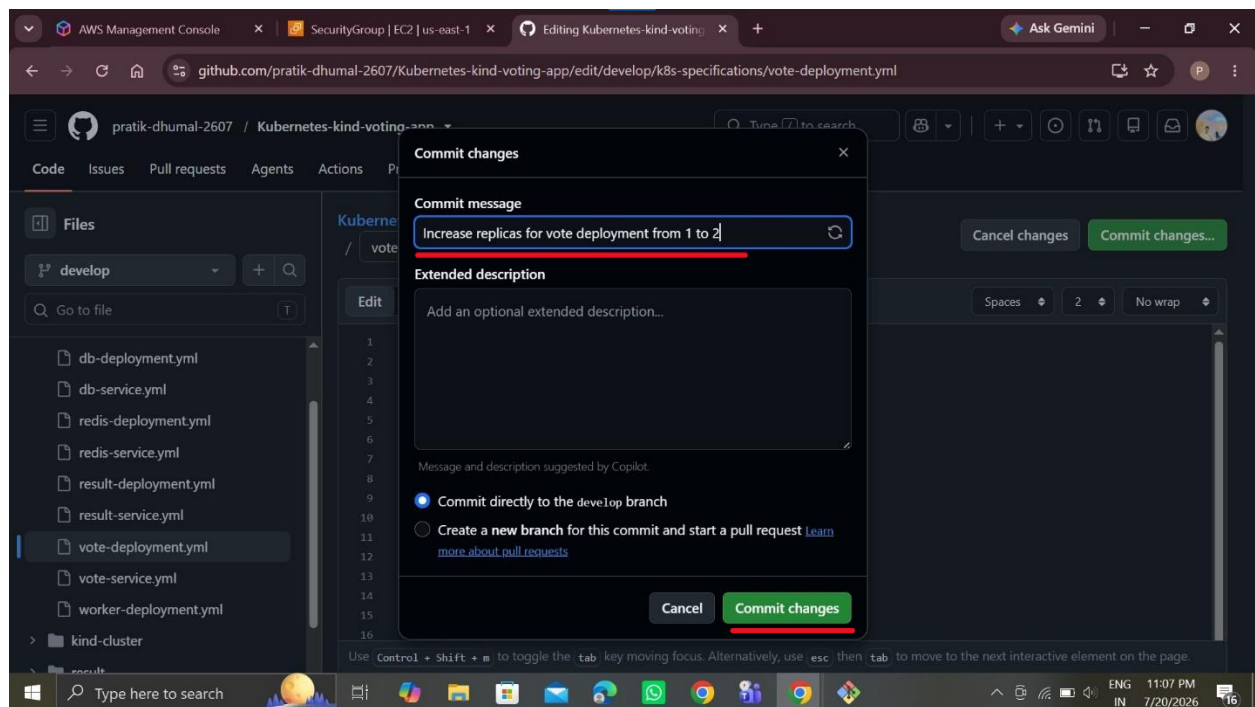
Then show this application in another format

The screenshot shows the Argo CD web interface in a browser window. The URL is `https://54.242.128.69:8443/applications/argocd/voting-app?view=network&resource=`. The left sidebar is identical to the previous screenshot. The main content area is titled 'APPLICATION DETAILS NETWORK' and includes the same tabs. The application status remains 'Healthy' and 'Synced to develop (945b96b)'. The last sync was successful 7 minutes ago. The main view displays a network diagram of the application's resources, showing the relationships between pods, services, and deployments.

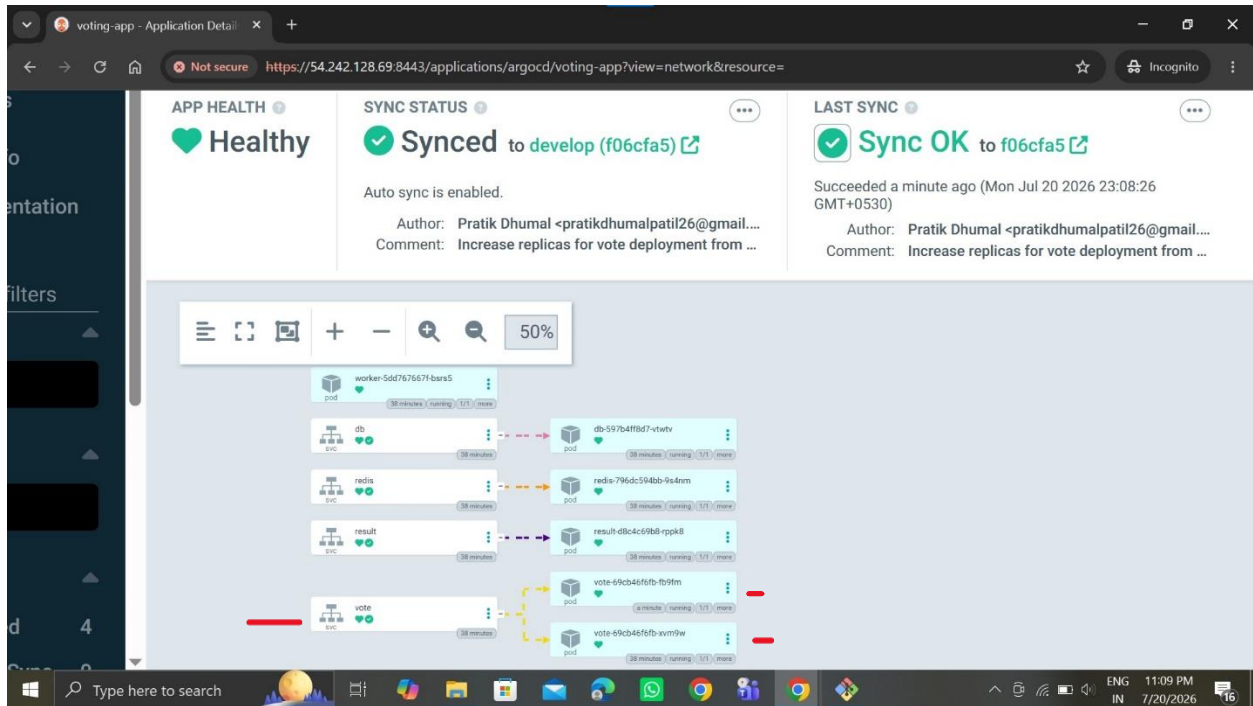
Then this pods show in terminal

```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
db-597b4ff8d7-vtwtv                1/1     Running   0           14m
redis-796dc594bb-9s4nm             1/1     Running   0           14m
result-d8c4c69b8-rppk8             1/1     Running   0           14m
vote-69cb46f6fb-xvm9w              1/1     Running   0           14m
worker-5dd767667f-bsrs5            1/1     Running   0           14m
ubuntu@ip-172-31-25-18:~/k8s-install$
```

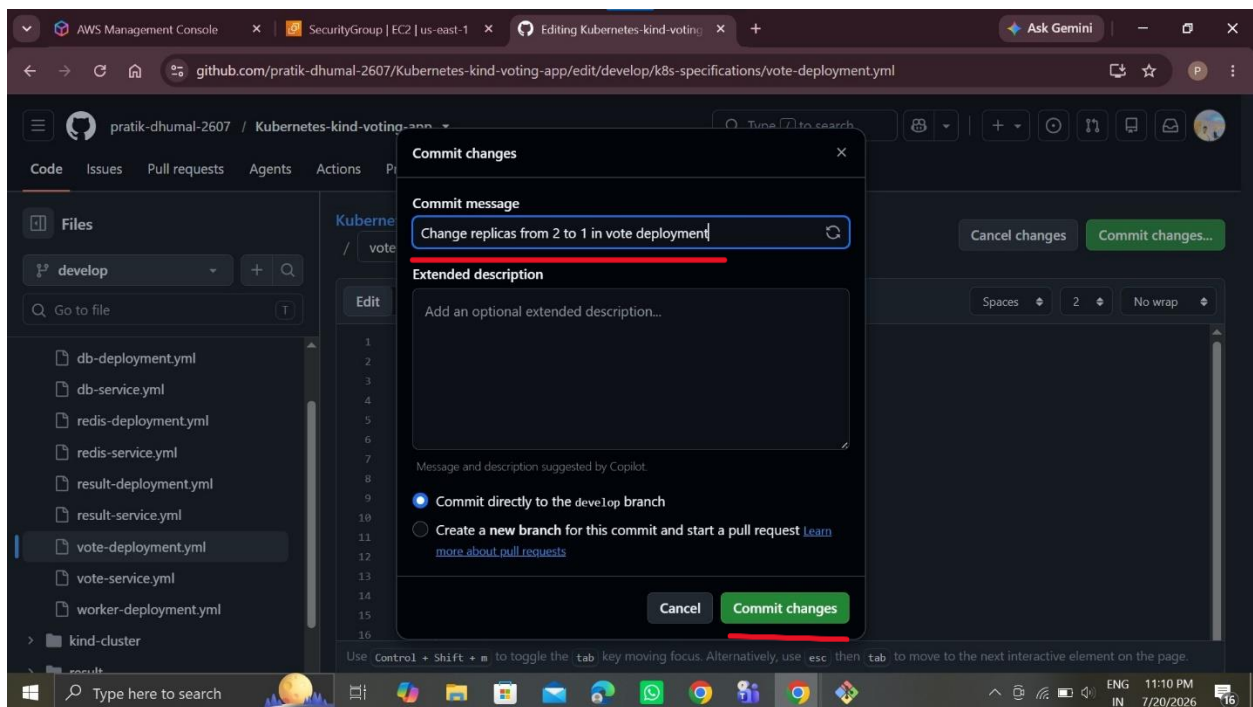
Then go to github & set vote replicas 1 to 2



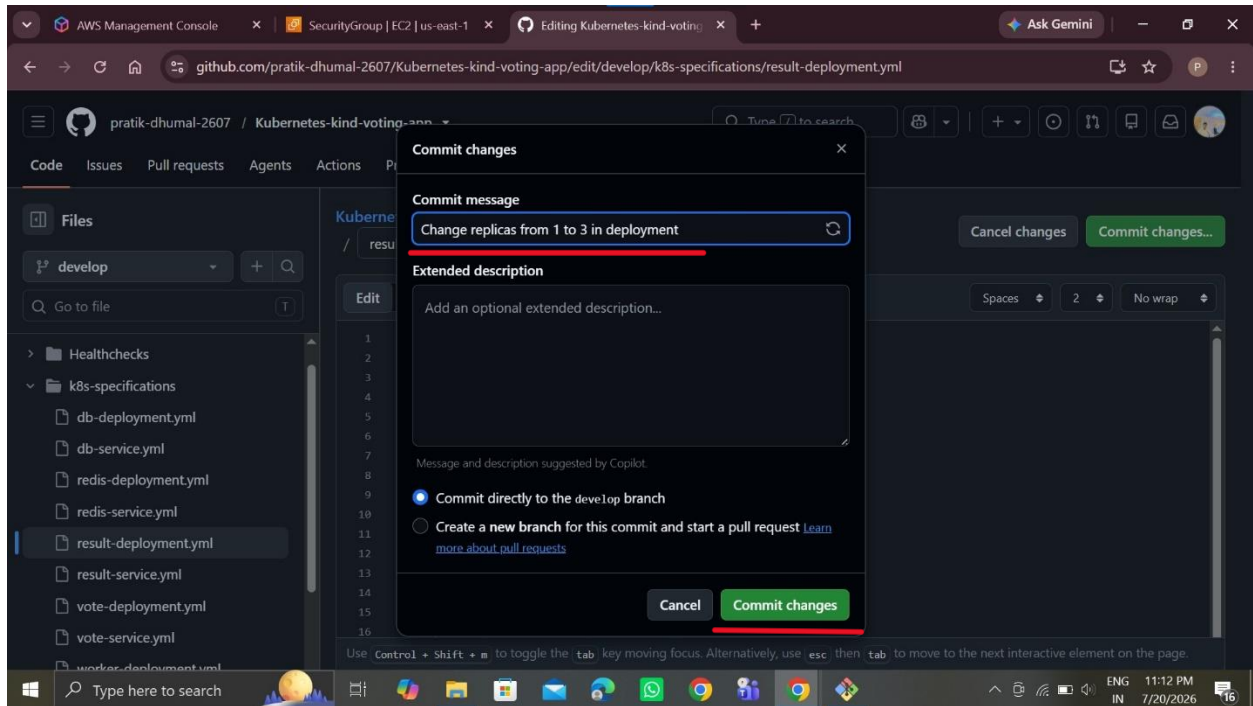
Successfully running 2 pods vote deployment



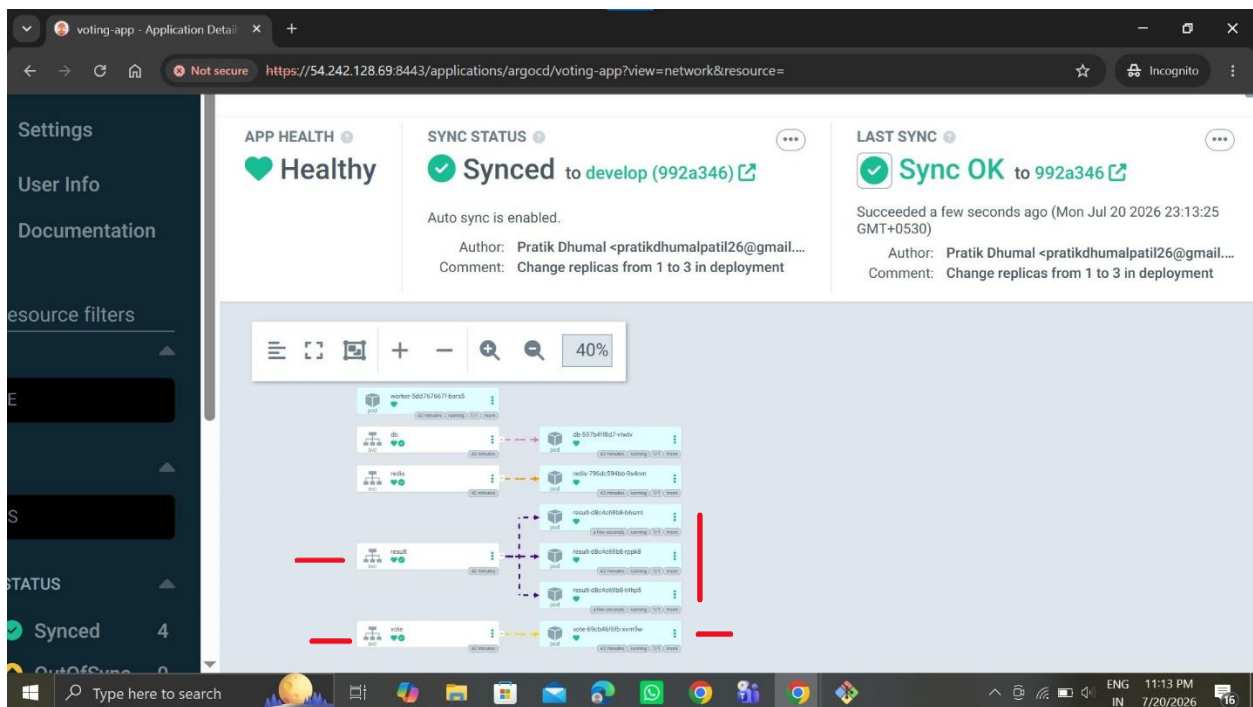
Then set vote replicas 2 to 1



Set result replicas 1 to 3



Successfully show this pods vote 1 pods running & result 3 pods running



```
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
db-597b4ff8d7-vtwtv	1/1	Running	0	40m
redis-796dc594bb-9s4nm	1/1	Running	0	40m
result-d8c4c69b8-rppk8	1/1	Running	0	40m
vote-69cb46f6fb-xvm9w	1/1	Running	0	40m
worker-5dd767667f-bsrs5	1/1	Running	0	40m

```
ubuntu@ip-172-31-25-18:~/k8s-install$  
ubuntu@ip-172-31-25-18:~/k8s-install$ Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
Handling connection for 8443  
  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
db-597b4ff8d7-vtwtv	1/1	Running	0	42m
redis-796dc594bb-9s4nm	1/1	Running	0	42m
result-d8c4c69b8-b6smt	1/1	Running	0	37s
result-d8c4c69b8-rppk8	1/1	Running	0	42m
result-d8c4c69b8-t4hp5	1/1	Running	0	37s
vote-69cb46f6fb-xvm9w	1/1	Running	0	42m
worker-5dd767667f-bsrs5	1/1	Running	0	42m

```
ubuntu@ip-172-31-25-18:~/k8s-install$ |
```

```

ubuntu@ip-172-31-25-18: ~/k8s-install
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443

ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
db-597b4ff8d7-vtwtv                1/1      Running   0           42m
redis-796dc594bb-9s4nm             1/1      Running   0           42m
result-d8c4c69b8-b6smt             1/1      Running   0           37s
result-d8c4c69b8-rppk8             1/1      Running   0           42m
result-d8c4c69b8-t4hp5             1/1      Running   0           37s
vote-69cb46f6fb-xvm9w              1/1      Running   0           42m
worker-5dd767667f-bsrs5            1/1      Running   0           42m

ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get deployments
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-2build1)
Try: sudo apt install <deb name>

ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get deployments
NAME    READY    UP-TO-DATE    AVAILABLE    AGE
db       1/1      1              1            44m
redis   1/1      1              1            44m
result  3/3      3              3            44m
vote    1/1      1              1            44m
worker  1/1      1              1            44m

ubuntu@ip-172-31-25-18:~/k8s-install$

```

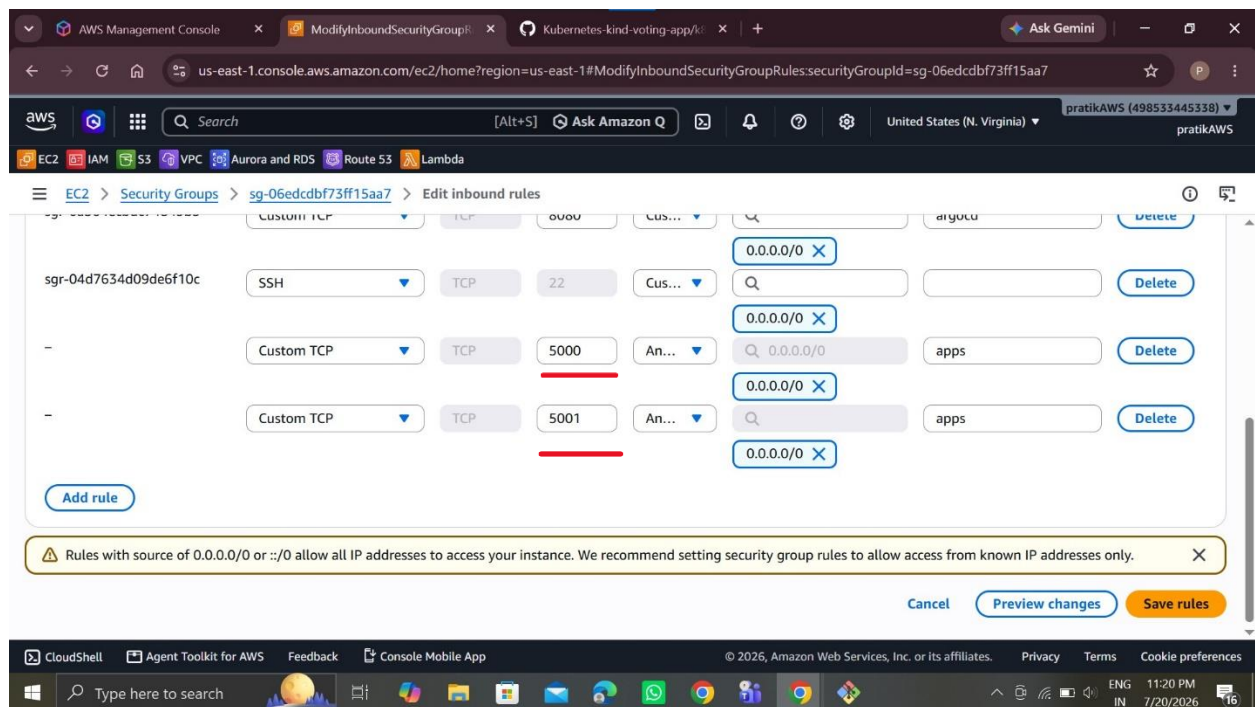
Port forward in vote & result

```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get svc
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
db            ClusterIP     10.96.236.193   <none>           5432/TCP         44m
kubernetes    ClusterIP     10.96.0.1       <none>           443/TCP          81m
redis         ClusterIP     10.96.34.237    <none>           6379/TCP         44m
result        NodePort      10.96.149.126   <none>           5001:31001/TCP   44m
vote          NodePort      10.96.253.142   <none>           5000:31002/TCP   44m
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl port-forward svc/vote 5000:5000 --address=0.0.0.0 &
[3] 13751
ubuntu@ip-172-31-25-18:~/k8s-install$ error: unknown command "port-forward" for "kubectl"

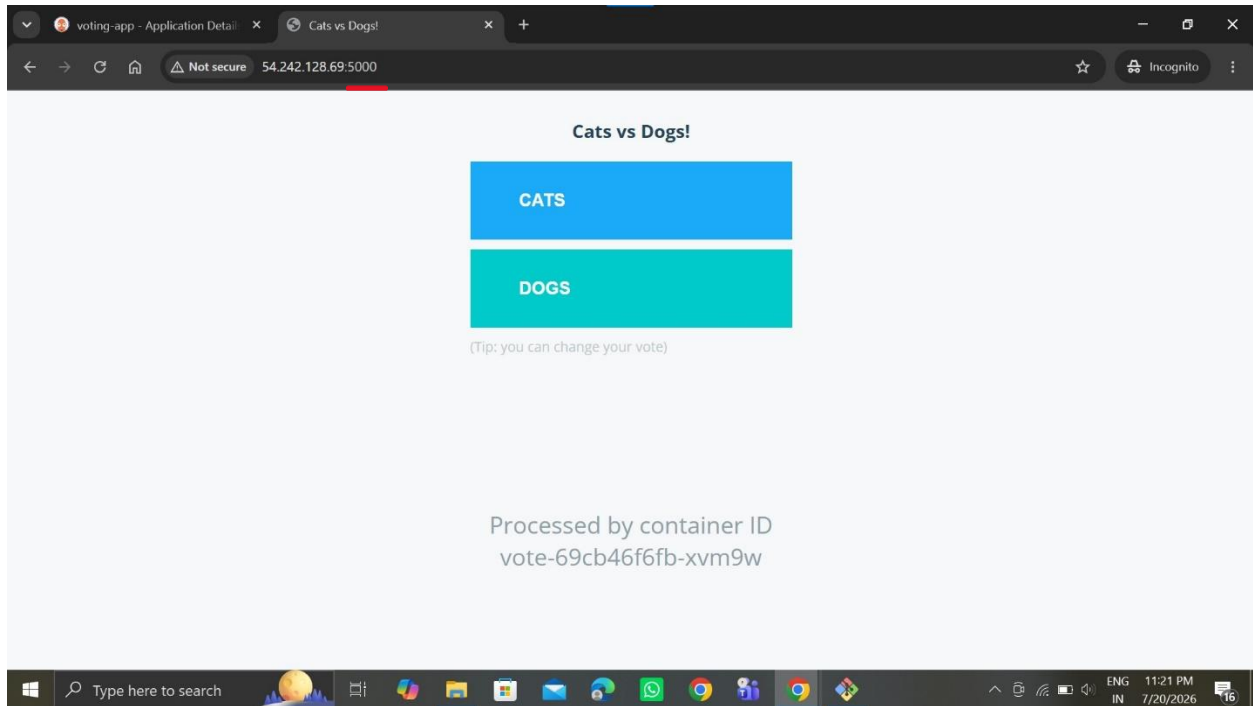
Did you mean this?
      port-forward
kubectl port-forward svc/vote 5000:5000 --address=0.0.0.0 &
[4] 13767
[3] Exit 1
      kubectl port-forward svc/vote 5000:5000 --address=0.0.0.0
ubuntu@ip-172-31-25-18:~/k8s-install$ Forwarding from 0.0.0.0:5000 -> 80

ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl port-forward svc/result 5001:5001 --address=0.0.0.0 &
[5] 13837
ubuntu@ip-172-31-25-18:~/k8s-install$ Forwarding from 0.0.0.0:5001 -> 80
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
```

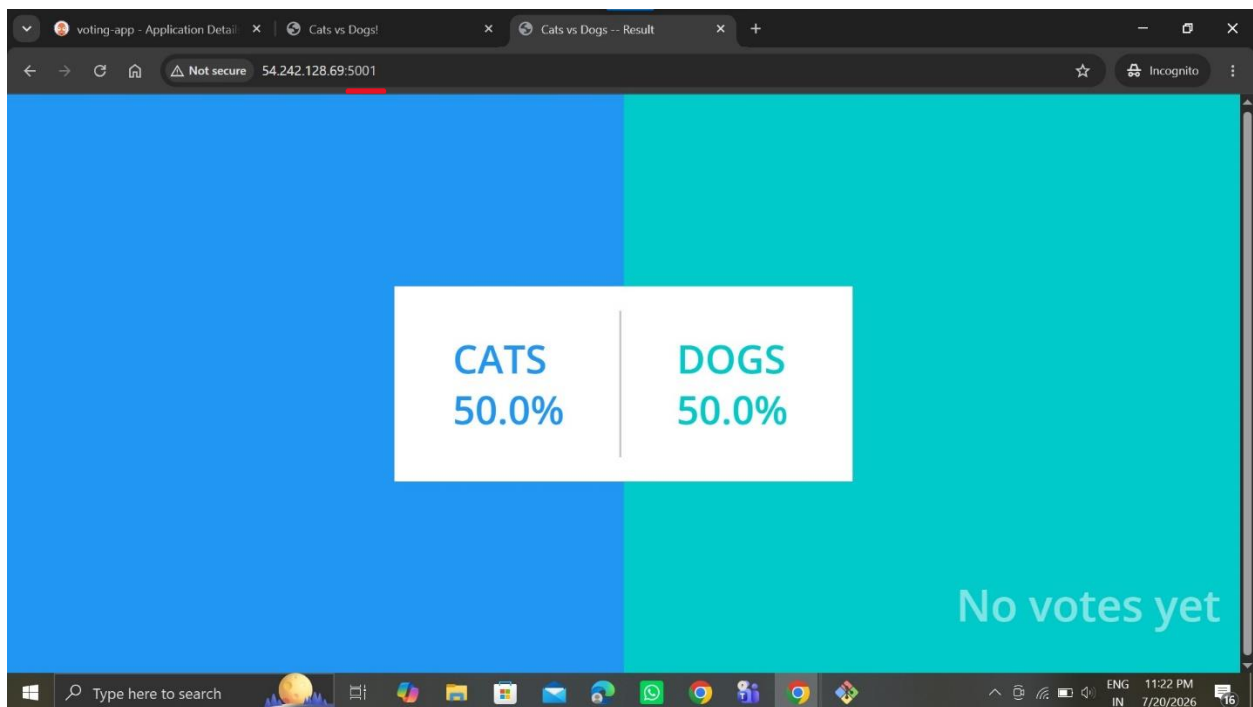
Allow this port in Security group



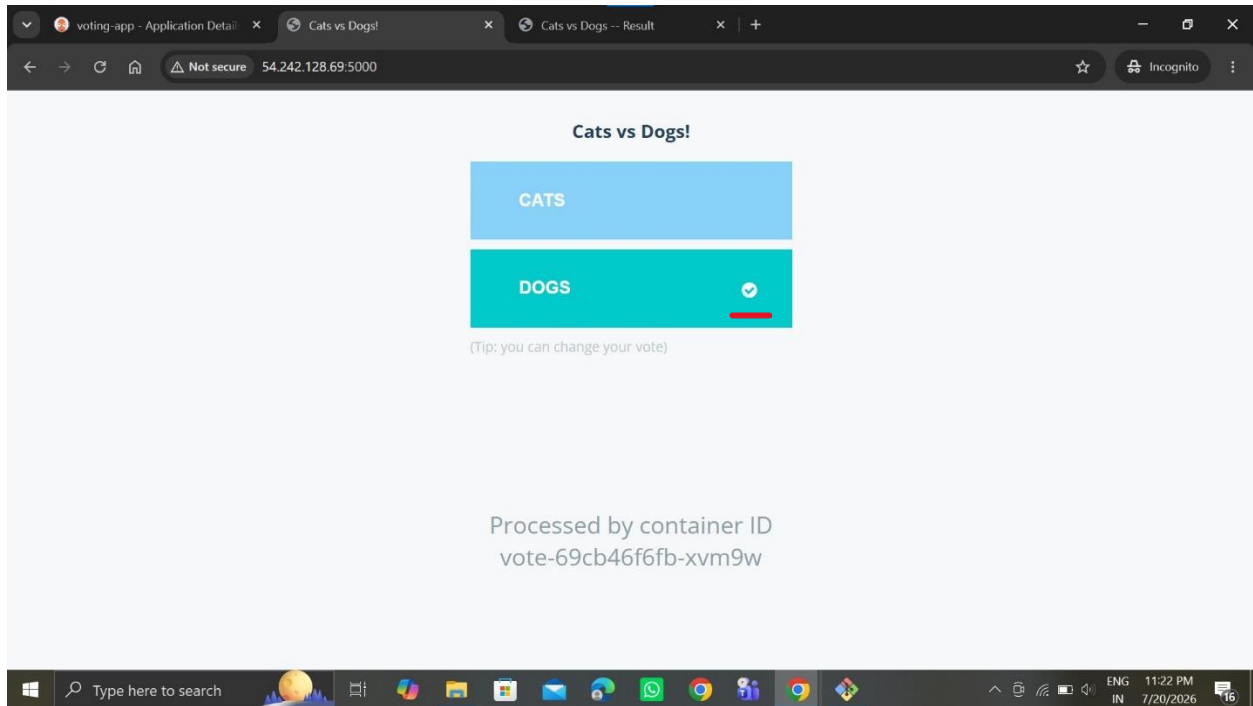
This is Vote page



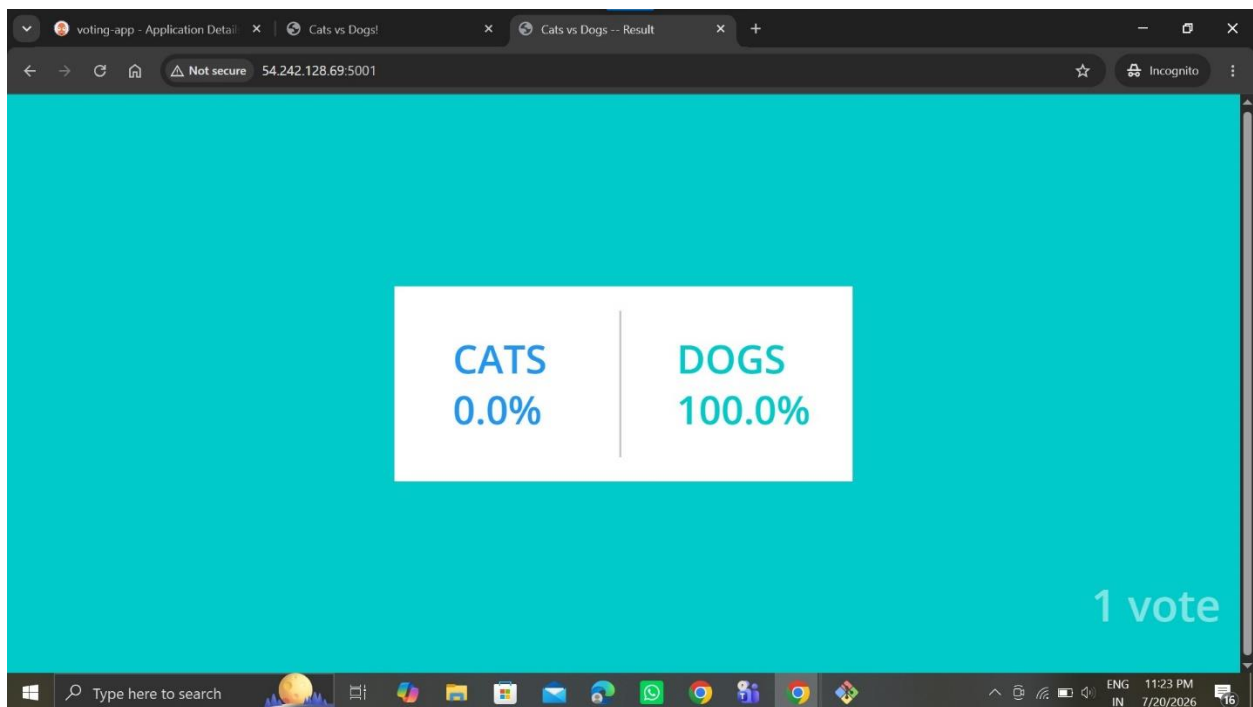
This is Result page



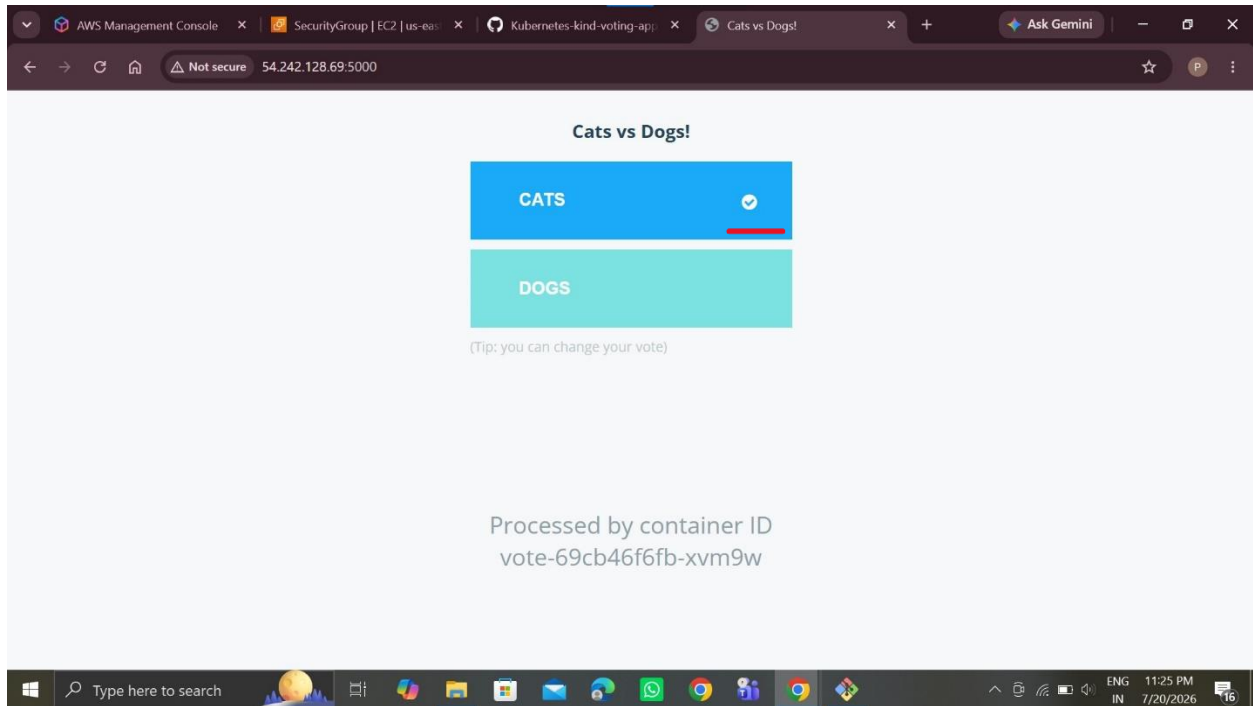
Then vote Dogs



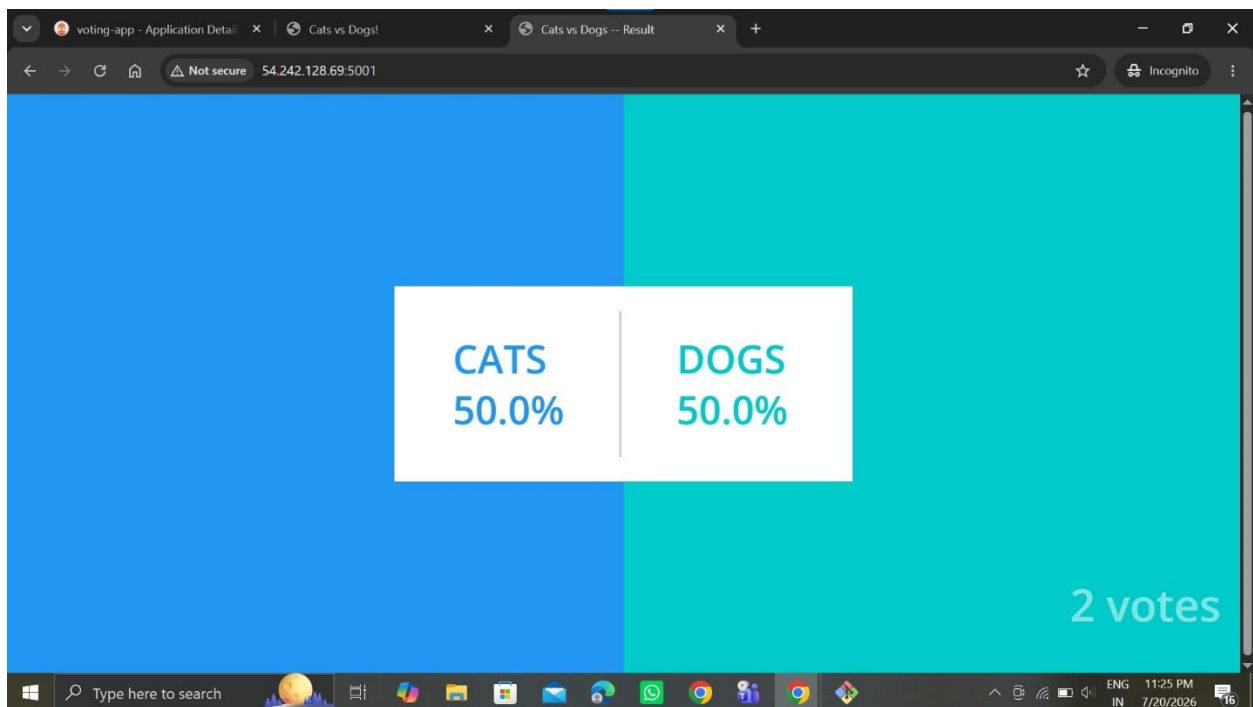
Result is 100% in Dogs



Then go to another browser tab and open vote page & vote to Cats



Then go to result page the result is equal



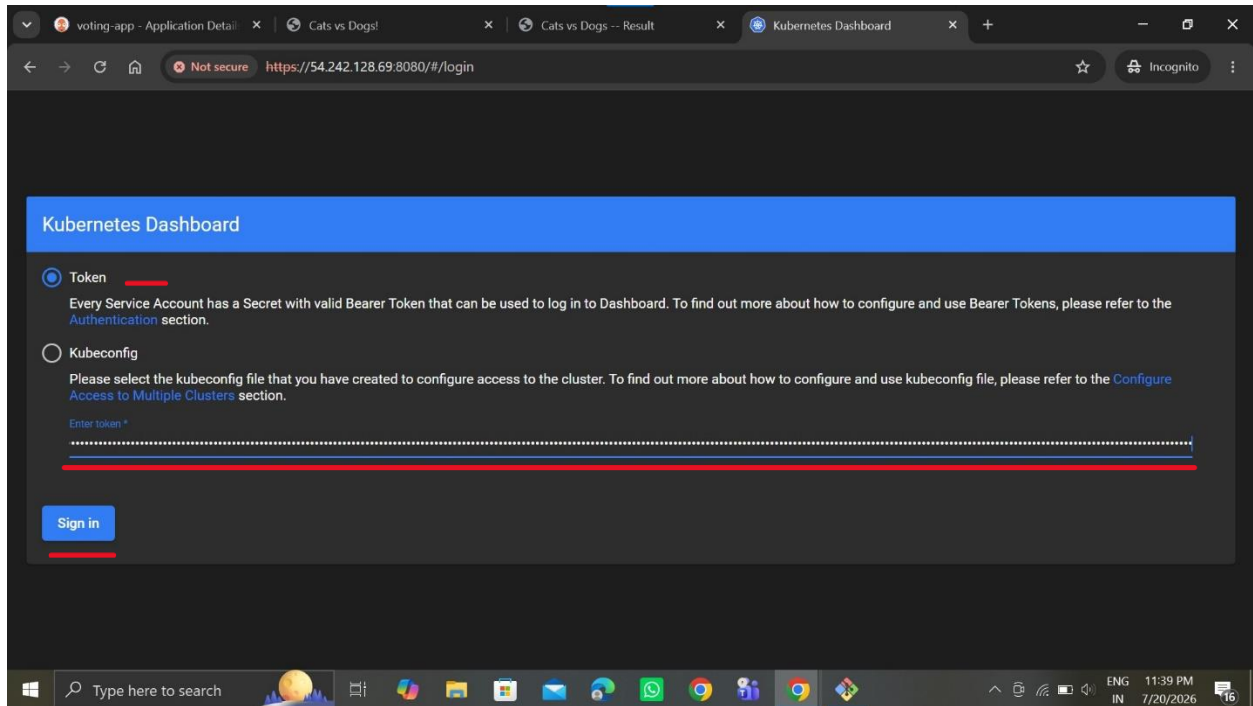
```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl apply -f https://raw.githubusercontent.com/kubernetes
/dashboard/v2.7.0/aio/deploy/recommended.yaml
namespace/kubernetes-dashboard created
serviceaccount/kubernetes-dashboard created
service/kubernetes-dashboard created
secret/kubernetes-dashboard-certs created
secret/kubernetes-dashboard-csrf created
secret/kubernetes-dashboard-key-holder created
configmap/kubernetes-dashboard-settings created
role.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrole.rbac.authorization.k8s.io/kubernetes-dashboard created
rolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
deployment.apps/kubernetes-dashboard created
service/dashboard-metrics-scraper created
deployment.apps/dashboard-metrics-scraper created
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ |
```

```
ubuntu@ip-172-31-25-18: ~/k8s-install$  
ubuntu@ip-172-31-25-18:~/k8s-install$ vim kubernetes-dashboard.yml  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl create namespace kubernetes-dashboard  
Error from server (AlreadyExists): namespaces "kubernetes-dashboard" already exists  
ubuntu@ip-172-31-25-18:~/k8s-install$  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl apply -f kubernetes-dashboard.yml  
serviceaccount/admin-user created  
clusterrolebinding.rbac.authorization.k8s.io/admin-user created  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl -n kubernetes-dashboard create token admin-user  
eyJhbGciOiJSUzI1NiIsImtpZCI6IjBwTmZSU09LSWc3cmc1bTFwQUlnaEkzLUZyRE80dDFDUVJv3bj1tCTB6SUV1fQ.eYjhdwqiOlsiaHR0CHM6Ly9rdWJlcml5IGVZLmRlZmF1bHouc3ZjLmNsdXN0ZXIubG9jYWwiXSwiZXhwIjojNXZgONTcOMTA1LCJpYXQiOjE3ODQ1NZA1MDUsImlycyI6Imh0dHBzoi8va3ViZXJuZXRlcy5kZWZhdx0LnN2Yy5jbHVzdGVyLmxvYyZFsiWiianRpIjoiyjY2OTU0ODctcnuyOC0ONjpmLWI1MQZtZTMzMzNmZGRjIiwia3ViZXJuZXRlcy5pbYi6eyJJyYw1lcm5lbGdzLWRhc2hib2FyzCisInN1cnpy2Vhy2NvdW50Ijp7Im5hbWUiOiJhZGlpbi1lc2VybmVkaWkiIjoidGd3MjFkgOGITmmNMky00Mzy3LEWEwZqtztZjmw5MTA1YjAxIn19LCJyYmYioiE3ODQ1NZA1MDUsInN1Yi6InN2Yy5kZXRlbnRpdzZjZjZm5ldGVZLmRhc2hib2FydDphZGIpb1l1c2VybmVudW50Ad_2h_cLKsfBER8fm7yk2wjht_nxqrvoixxxEowBsm3CsorPFROYGanhgzeeek2-wx2-VHgqOlVOJN9rfcdLqim-BQwyby2G7GA8wn6ucSGzaotoG64GCGlmInXG-qm8dn5HRV57B0TMzmd9pb9xzS4w_18K_4ql04865GDtiKBi6BAEGYoVRNA_HHDcb8s9rwtc_1njmcBMCCs-mFiL5xcjWHKRd9-iVzlufzzxuWDW_zf9RDhs5yx FJ2bu9dv66grhtRqvfvzv7j8o_wDMsimtBVtMcclYkze6I2PgoBMeotDwdZqd-JbPMvxAlrxMYPNlRbijrq
```

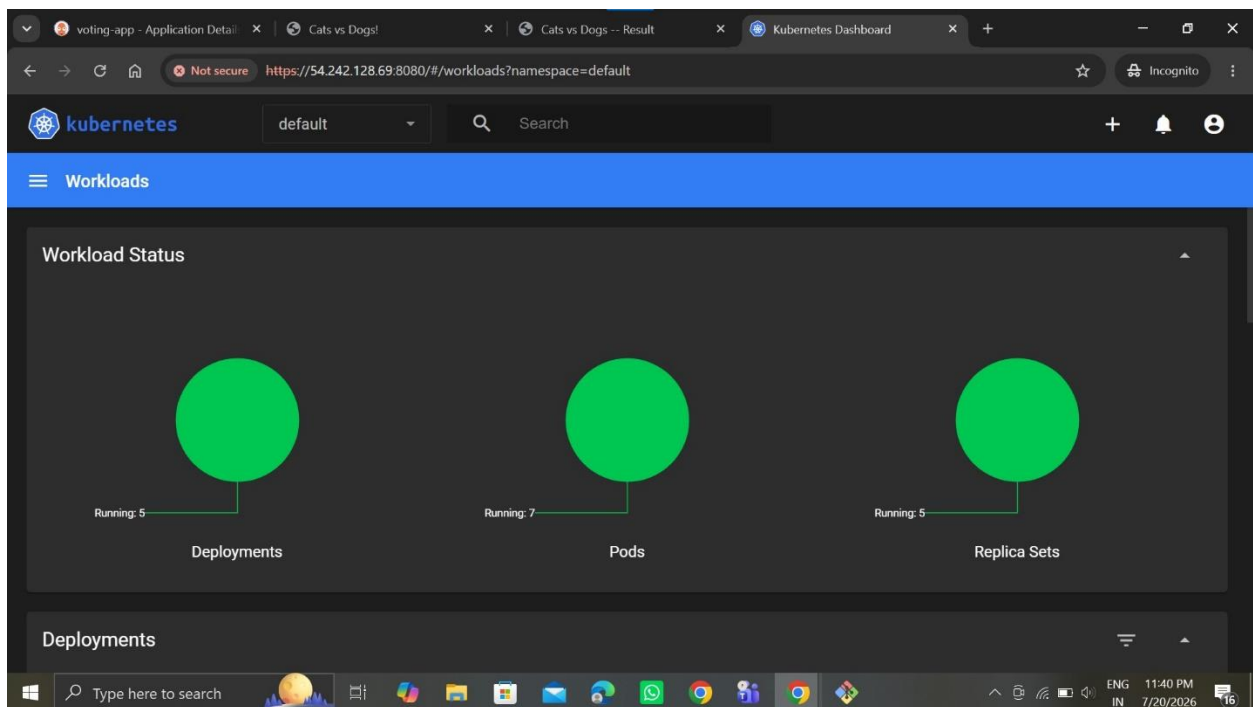


```
ubuntu@ip-172-31-25-18:~/k8s-install$  
zS4w_18K_4qL04865GDTiK6iBAgEgyOYrNa_HHDCb8s9rWtc_1njmcbMCCcs-mFiL5xcjWHKRd9-ivZluFzzxuWDW_Zf9RDhs5yx  
FJ2bu9dv66qrhtRqvzfz7j8o_WDMsimtBVTMcC1yke6I2PgoBMeoTdwZqtD-JbPMVxa1rxMPYNlRbjrq  
ubuntu@ip-172-31-25-18:~/k8s-install$  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get svc -n kubenetes-dashboard  
No resources found in kubenetes-dashboard namespace.  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get svc -n kubernet-es-dashboard  
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP    PORT(S)          AGE  
dashboard-metrics-scraper           ClusterIP           10.96.178.78     <none>         8000/TCP         6m51s  
kubernet-es-dashboard               ClusterIP           10.96.3.57       <none>         443/TCP          6m51s  
ubuntu@ip-172-31-25-18:~/k8s-install$  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl port-forward svc/kubernet-es-dashboard -n kubernet-es-d  
ashboard 8080:443 --address=0.0.0.0 &  
[6] 14955  
ubuntu@ip-172-31-25-18:~/k8s-install$ Forwarding from 0.0.0.0:8080 -> 8443  
  
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl -n kubernet-es-dashboard create token admin-user  
eyJhbGciOiJIUzI1NiIsImtpczkiOiJBTWVzdXNhc1swc3cmc1bTFwQUNaEkZLUZYRE80dDFCUlV3bjI1CTB6SUUiFiQ.eYjhDwoi  
o1siaHR0ciHM6ly9drJlcm51ldGVzLmRlZmFlbnQuc3ZjLmNsSDN0ZXIubG9jYWwyZSwiZWxhIjoxnZgoONDA5LjpxYXQioJE  
3ODQlNZA4MDksInlzcyl6Imh0dHBz0i8va3ViZXZXRlcyc5KZWZhdHwxOlNn2Vy5jbHVzdGvylmxvy2FsIiwianRpIjoijnjrd  
QlYzqtZDEzSO0YTfhLTk2OGMtNGhiWTNjYzY4OGUxiIiwia3ViZXZXRlcyc5pbYi6eyjuYw1lc3Bhy2UioijrdwJlcm5ldGVzL  
WRhc2hib2ZyZCisInNlcnZpY2VhY2NdvdW50Ijp7Im5hbWUiOiJhZGlpbi1lc2VyIiwidwklIjoimDG3mjFkoGItMmNmKy00MzY3  
LEWewZjtZTZjMWU5MTA1YjAxInhl9CLjUyMyYoje3ODQlNZA4MDksInNlcnZlI6InN5c3RlbtpzZXJ2aWNlYWNjb3VudDprdwJlcm5  
ldGVzLmRhc2hib2ZyZDphXzIxpb1lc2VyIn0.C8TT4whCEaeCOVFhkwc8UmK3kuQjBiTGO-_jcpPnzB-K-_yuc7vdbQfyIX4ddx  
9KVtd-CAWCVFBes4f2q6hxTx7QEz2EZTEeqgc960vwDCFfwynzPMHbkOU-qJTTLZjULNzhgor5my0jiZhHWB_OHyH8-jitUJRle-  
wn6RoBBZe6nmichRM2zs_bK3N1LJXlFIQpnIFbdPLwkLLgtSPZj16ySnY_7635akmtvqq7grQZ7U3Bjp_Q3FblYscQJ-Xzw9S9cm  
96vtenu6gLkv4a6_N7NwpwoygMYT1tzX805o6tlbmckoEntdf7LCqyQvUEaqfqlof5w9wwc9K3hx4GUug  
ubuntu@ip-172-31-25-18:~/k8s-install$  
ubuntu@ip-172-31-25-18:~/k8s-install$ |
```

Then paste this generated Token & sign in



Successfully show the Kubernetes Dashboard



Deployments pods

The screenshot shows the Kubernetes Dashboard interface. The top navigation bar includes the Kubernetes logo, a namespace dropdown set to 'default', a search bar, and user profile icons. The main content area is titled 'Workloads' and contains two sections: 'Deployments' and 'Pods'.

Deployments Table:

Name	Images	Labels	Pods	Created ↑
db	postgres:15-alpine	app: db	1 / 1	an hour ago
redis	redis:alpine	app: redis	1 / 1	an hour ago
result	dockersamples/examplevotingapp_result	app: result	3 / 3	an hour ago
vote	dockersamples/examplevotingapp_vote	app: vote	1 / 1	an hour ago
worker	dockersamples/examplevotingapp_worker	app: worker	1 / 1	an hour ago

Pods Table:

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created ↑
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The bottom of the dashboard shows a Windows taskbar with various application icons and system tray information including the date and time (11:40 PM 7/20/2026).

Pods

This screenshot shows the 'Pods' page within the Kubernetes Dashboard. The left sidebar contains a navigation menu with options like 'Workloads', 'Cron Jobs', 'Daemon Sets', 'Deployments', 'Jobs', 'Pods', 'Replica Sets', 'Replication Controllers', 'Stateful Sets', 'Service', 'Ingresses', 'Ingress Classes', and 'Services'. The 'Pods' option is currently selected.

The main content area displays a table of pods:

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)
result-d8c4c69b8-b6smt	dockersamples/examplevotingapp_result	app: result pod-template-hash: d8c4c69b8	my-cluster-worker	Running	0	-	-
result-d8c4c69b8-t4hp5	dockersamples/examplevotingapp_result	app: result pod-template-hash: d8c4c69b8	my-cluster-worker2	Running	0	-	-
db-597b4ff8d7-vlwtv	postgres:15-alpine	app: db pod-template-hash: 597b4ff8d7	my-cluster-worker2	Running	0	-	-
redis-796dc594bb-9s4nm	redis:alpine	app: redis pod-template-hash: 796dc594bb	my-cluster-worker2	Running	0	-	-

The bottom of the dashboard shows a Windows taskbar with various application icons and system tray information including the date and time (11:43 PM 7/20/2026).

Pods									
Replica Sets	●	result-d8c4c69b8-rppk8	dockersamples/examplevotingapp_result	app: result	pod-template-hash: d8c4c69b8	my-cluster-worker	Running	0	-
Replication Controllers									
Stateful Sets									
Service	●	vote-69cb46f6b-xvm9w	dockersamples/examplevotingapp_vote	app: vote	pod-template-hash: 69cb46f6b	my-cluster-worker	Running	0	-
Ingresses									
Ingress Classes									
Services	●	worker-5dd767667f-bsrs5	dockersamples/examplevotingapp_worker	app: worker	pod-template-hash: 5dd767667f	my-cluster-worker	Running	0	-

Replica sets

voting-app - Application Details					Cats vs Dogs!					Cats vs Dogs -- Result					Kubernetes Dashboard				
← → ↻ ↺					Not secure https://54.242.128.69:8080/#/workloads?namespace=default					☆ Incognito									
kubernetes					default					Search					+ 🔔 👤				
☰ Workloads																			
Replica Sets																			
Name		Images		Labels		Pods		Created ↑											
● db-597b4ff8d7		postgres:15-alpine		app: db pod-template-hash: 597b4ff8d7		1 / 1		an hour ago			⋮								
● redis-796dc594bb		redis:alpine		app: redis pod-template-hash: 796dc594bb		1 / 1		an hour ago			⋮								
● result-d8c4c69b8		dockersamples/examplevotingapp_result		app: result pod-template-hash: d8c4c69b8		3 / 3		an hour ago			⋮								
● vote-69cb46f6b		dockersamples/examplevotingapp_vote		app: vote pod-template-hash: 69cb46f6b		1 / 1		an hour ago			⋮								
● worker-5dd767667f		dockersamples/examplevotingapp_worker		app: worker pod-template-hash: 5dd767667f		1 / 1		an hour ago			⋮								

Automated Deployment of Scalable Applications on AWS EC2 with Kubernetes and Argo CD Successfully .